# Effective Anomaly Space for Hyperspectral Anomaly Detection

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***Abstract*— Due to unavailability of prior knowledge about anomalies, background suppression (BS) is a crucial factor in anomaly detection (AD) evaluation. The difficulty in dealing with BS arises from the fact that anomalies are generally sandwiched between background (BKG) and noise. This article presents a new concept, called effective anomaly space (EAS), to resolve this dilemma. To accomplish this goal, the well-known independent component analysis (ICA) is used to address between BKG and anomalies issue by removing the first two orders of data statistics (2OS), while sparsity cardinality (SC) is used to address between anomalies and noise issue by removing non-Gaussian noises and interferers from anomalies. Specifically, SC is rederived as fixed SC (FSC) corresponding to fixed length coding for a spectral vector and a spatial band and variable SC (VSC) for a corresponding to variable length coding for a spectral–spatial sample and spatial–spectral band from an information theory point of view. Combining ICA and the new derived SC versions allows EAS not only to remove BKG-characterized by 2OS including Gaussian-distributed signal sources but also to remove non-Gaussian noises/interferers from anomalies. As a consequence, EAS can significantly increase anomaly detectability. In particular, one of great benefits resulting from EAS is that EAS can also improve current low-rank and sparse representation (LRaSR)-based methods used for AD.**

***Index Terms*— Anomaly space (AS), data sphering (DS), effective AS (EAS), independent component analysis (ICA), sparsity cardinality (SC), virtual dimensionality (VD).**

LIST OF ACRONYMS

| | |
|---|---|
| 2OS | second-order statistics |
| AD | anomaly detection |
| AS | anomaly space |
| ATGP | automatic target generation process |
| AUC | area under an ROC curve |
| BKG | background |
| BS | background suppression |
| CEM | constrained energy minimization |
| CEM-AD | CEM anomaly detector |
| CDA | component decomposition analysis |
| CRD | collaborative representation detector |
| EAS | effective anomaly space |
| ECEM | ensemble cascaded CEM |
| FastICA | fast independent component analysis |
| FSC | fixed sparsity cardinality |
| GoDec | go decomposition |
| hCEM | hierarchical CEM |
| HFC | Harsanyi-Farrand-Chang |
| HOS | high-order statistics |
| IC | independent component |
| ICA | independent component analysis |
| ICSC | independent component sparsity cardinality |
| $j$-FSSC | $j$-fixed spectral sparsity cardinality |
| $j$-BSSC | $j$-fixed band sparsity cardinality |
| $jL$-VSC | spatial–spectral band variable sparsity cardinality |
| $jN$-VSC | spectral–spatial sample variable sparsity cardinality |
| KCEM | kernel CEM |
| LRaSMD | low-rank and sparse matrix decomposition |
| LRR | low-rank representation |
| LSDM | low-rank and sparse decomposition model |
| MD | Manhattan distance |
| MoG | mixture of Gaussians |
| MXSVD | minimax singular value decomposition |
| NWHFC | noise-whiten HFC |
| ODP | overall detection performance |
| OSP | orthogonal subspace projection |
| RPCA | robust principal component analysis |
| PTA | prior-based tensor approximation |
| RX-AD | Reed–Xiaoli anomaly detector |
| ROC | receiver operating characteristic |
| SC | sparsity cardinality |
| SNPR | signal-to-noise probability ratio |
| TD | target detectability |
| VD | virtual dimensionality |
| VSC | variable sparsity cardinality |

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## I. INTRODUCTION

ONE of the fundamental tasks in hyperspectral imaging is anomaly detection (AD) because anomalies cannot be known *a priori* but provide crucial and vital information for image analysis [1]–[5]. It is very challenging in the sense that anomalies generally possess the following unique properties. First, anomalies usually cannot be identified by visual inspection or prior knowledge. Second, the presence of anomalies is unexpected. Third, the probability of occurrence of anomalies is relatively low. Fourth, once anomalies occur, their populations cannot be too many. Most importantly, due to the nature of anomalies, the number of anomalies present in the data will be very limited, in which case anomalies cannot constitute Gaussian distributions by statistics.

In order for an anomaly detector to perform well, two factors play a key role in effectiveness of AD, background (BKG) suppressibility, and anomaly detectability. As an illustrative example, we consider the well-known anomaly detector developed

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by Reed and Xiaoli [6], referred to as Reed–Xiaoli anomaly detector (RX-AD) given by

$$\delta^{\text{RX-AD}}(\mathbf{r}) = (\mathbf{r}-\boldsymbol{\mu})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}) \tag{1}$$

where $\mathbf{r}$, $\boldsymbol{\mu}$, and $\mathbf{K}$ are the data sample vector, global sample mean vector, and global sample covariance matrix, respectively. According to (1), there are two processes involved in RX-AD. One is BKG suppression (BS) by inverting the covariance matrix $\mathbf{K}$ via $\mathbf{K}^{-1}$. The other is detection of anomalies by a matched filter using the matching signature $(\mathbf{r}-\boldsymbol{\mu})$ via an inner product by $(\mathbf{r}-\boldsymbol{\mu})^T$.

Over the past years, many AD techniques have been taking approaches by dealing with the first issue, i.e., how to suppress or eliminate BKG effect to improve AD. Earlier examples are to manipulate the sample covariance matrix $\mathbf{K}$ for this purpose. Of particular interest is [7], [8] which replaced $(\mathbf{r}-\boldsymbol{\mu})$ and $\mathbf{K}$ in (1) with $\mathbf{r}$ and the global sample correlation matrix $\mathbf{R}$ to derive a variant of RX-AD, known as R-AD, given by

$$\delta^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}. \tag{2}$$

Interestingly, Chang further showed in [9] that R-AD is indeed a special case of the constrained energy minimization (CEM) anomaly detector (CEM-AD). Other examples include using local sample covariance/correlation matrices such as local RX-AD [10], [11], support vector data description (SVDD) [12], weighted-RX-AD and linear filter-based RX-AD in [13], and a sliding window [14]; a dual-window-based eigen separation transform (DWEST) was developed in [15] followed by collaborative representation-based detector (CRD) in [16] which also used dual windows; and multiple-window anomaly detection (MWAD) by making use of multiple windows to perform AD adaptively in [17], guided-filtering-based anomaly detector in [18], spectral–spatial feature-extraction-based anomaly detector in [19], and a nonlinear anomaly detector, kernel-based anomaly detector (KRX-AD), in [20]. In addition, there were also anomaly detectors developed for various applications such as unlabeled classification in [21] and real-time processing of AD in [22].

There are also approaches that deviated from the above-mentioned traditional extensions, by focusing on separating, estimating, reconstructing BKG from the data space, and then performing AD on the residual data space. Examples include BKG separation in [23], autoencoders for BKG reconstruction in [24], BKG estimation in [25], deep learning in [26], low-rank and sparse representation (LRaSR) in [27], low-rank and sparse matrix decomposition in [28]–[30], and BKG anomaly tensor decomposition in [31]. While such approaches can achieve success to some extent, a crucial issue has been overlooked, that is, enhancement of anomaly detectability.

In recent years, the issue of how to enhance AD was addressed by operating RX-AD in two separate spaces instead of the same data space as commonly anomaly detectors do such as RX-AD in (1) and R-AD in (2). Specifically, RX-AD in (2) can be further modified and extended in [32] to

$$\delta^{\text{RX-AD}}(\mathbf{r_A}) = (\mathbf{r_A}-\boldsymbol{\mu_B})^T\mathbf{K_B}^{-1}(\mathbf{r_A}-\boldsymbol{\mu_B}) \tag{3}$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the sparse space and low-rank space, respectively. In this case, RX-AD uses $\mathbf{K_B}^{-1}$ to suppress BKG in the space $\mathbf{B}$, while using $(\mathbf{r_A}-\boldsymbol{\mu_B})$ to enhance AD in the space $\mathbf{A}$. As shown by experiments in [32], the RX-AD specified by (3) significantly improved RX-AD in AD. It is also interesting to note that low-rank and sparse matrix decomposition (LRaSMD) developed by go decomposition (GoDec) in [29] also decomposed the data space into three separate spaces with a low-rank space representing the BKG space, a sparse space retaining anomalies, and a noise space. With noises being separated from the sparse space, LRaSMD has been shown in [32] to outperform robust principal component analysis (RPCA) and other LRaSR-based approaches. In [33], Chen *et al.* further devised a new model, called component decomposition analysis (CDA), to decompose the data space into principal component (PC) space, independent component (IC) space, and noise component space to show that CDA implemented in conjunction with GoDec (CDASC) indeed had advantages over LRaSR and RPCA by incorporating the concept of sparsity cardinality (SC) introduced by GoDec to improve AD performance. This CDASC explains the reason why RPCA and LRaSR did not work well in general simply because noise sources cannot be effectively separated from their sparse spaces.

Despite that CDASC can separate noises from a sparse space, there are still issues arising from the fact that anomalies are indeed sandwiched between BKG space and noise space. When anomalies are strong in terms of the second-order statistics, they can be easily considered as BKG signatures. At the other extreme, when anomalies are weak, these anomalies may be mistakenly treated as noisy signal sources. These two issues are in fact closely related to the BS issue detailed as follows.

1) AD in BKG to address between BKG and anomaly issue: It assumes that BKG can be characterized by the first two orders of data statistics (2OS), while anomalies can be specified by high orders of data statistics (HOS). In this case, we can use data sphering (DS) to remove 2OS. It is worth noting that DS is very different from LRaSR and LRaSMD model-based approaches since low-rank space may not be completely characterized by 2OS. Consequently, it may be very likely that the BKG signatures characterized by 2OS will be still leaked into the sparse space to be considered as anomalies. There are also other approaches which used a mixture of Gaussian distributions (MoGs) in [25] to model the BKG space. Unfortunately, this is not effective because non-Gaussian noises/interferers cannot be characterized by 2OS and thus cannot be removed by MoG due to the fact that MoG is still Gaussian. Besides, the number of Gaussian distributions used for MoG is crucial. If MoG is estimated incorrectly, then BKG cannot be well-characterized, in which case certain BKG signatures that cannot be described by MoG will be still considered as anomalies. Therefore, in either case, anomalies are still contaminated by BKG signatures. So, using DS to remove 2OS is a better means than using a mixture of Gaussians to remove BKG. In other words, the use of DS can improve the clean-up of BKG which cannot be characterized by Gaussian statistics.

2) AD in noises/interferers to address between anomalies and noise/interferers issue:

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Due to a small number of anomalies present in the data, anomalies are generally assumed to be characterized by HOS as opposed to BKG which is characterized by 2OS. Accordingly, anomalies can be considered as HOS targets. It is known that independent component analysis (ICA) [34] is an HOS technique which has been widely used for blind source separation in signal processing, speech processing, and medical imaging. This implies that ICA can unmix anomalies via its generated statistically ICs. Unfortunately, one inherent issue arising in ICA is its inability in unmixing non-Gaussian noises/interferers via ICs. This indicates that non-Gaussian noises/interferers can also be very likely to be treated as anomalies. If we consider anomalies are embedded in such non-Gaussian noises, then how to detect anomalies in non-Gaussian noises/interferers still remains unsolved. Interestingly, we can consider the space formed by all the ICA-generated ICs as HOS space (this is because we have removed 2OS and assume that anomalies can be characterized by HOS). Then we further divide the HOS space into two disjoint component spaces, one to accommodate anomalies, referred to as anomaly space (AS), and the other to accommodate non-Gaussian noises and interferers, referred to as anomaly BKG (A-BKG) space which can be considered as BKG of anomalies. In this case, AD in BKG can take place by constraining an anomaly detector in AS, while suppressing the A-BKG space. This practice is similar to LRaSMD and performs essentially BS for AS by operating an anomaly detector in the sparse space with A-BKG space being separated by the noise space. However, such idea seems to be easier said than done because three issues need to be resolved. One is the use of random initial conditions to generate ICs, which leads to nonrepeatable and inconsistent ICs. A second one is how many ICs are needed to construct an AS. A third one is how to take care of non-Gaussian noises/interferers in IC components.

As for the first issue, an approach, FastICA developed in [35]–[37], with its algorithm available on website http://research.ics.aalto.fi/ica/fastica/ can be used to generate a sequence of ICs but its use of random initial conditions makes its generated ICs not repeatable and inconsistent. To resolve this issue, an automatic target generation process (ATGP) developed in [38] can be used to generate a desired set of initial projection vectors to be used by FastICA so that each ATGP-generated target can be used as the desired initial condition for each IC generation process. Since the ATGP-generated targets are deterministic, the FastICA generated ICs are also deterministic and repeatable. The resultant FastICA is referred to as ATGP-FastICA [39].

As for the second issue, the virtual dimensionality (VD) in [8] and [40]–[44] coupled with a minimax-singular value decomposition (MXSVD) developed in [45] can be jointly used to determine a desired number of ICs, $j$, such that the first $j$ ICs produced by ATGP-FastICA can be used to form an IC component space as AS.

Finally, as for the third issue, SC introduced by GoDec [29] is further imposed on AS to construct an effective AS (EAS). Its idea is derived from an assumption that noise is random and ubiquitous over AS. So, noise data sample values across the entire spectral range are generally nonzero and also relatively small. This implies that noises in AS cannot be sparse. In this case, if the value of SC, $k$ is imposed on AS, and then we can retain those data sample values with the first $k$ nonzero values in AS, while setting the remaining data sample values to zeros. Such obtained SC-constrained data samples can be considered as potential anomalous data samples to form an EAS, which can actually remove non-sparse and non-Gaussian noises/interferers from AS. The key issue is how to determine an appropriate value of $k$ which was determined empirically in [29]. Although Zhang *et al.* [30] suggested that $k = j \times N$ where $N$ is the total number of data samples, they did not explain how and why $k$ was selected. In EAS, the concept of SC is rederived in a quite different way using source coding in information theory [46]. More specifically, the original idea of SC can be explored from two perspectives in terms of fixed sparsity cardinality (FSC) and variable sparsity cardinality (VSC). For a given value of $j$ which is the number of ICs, total number of data samples, $N$ and total number of spectral bands $L$, two versions of FSC can be defined from a viewpoint of fixed length coding in [46], that is, $j$-fixed spectral sparsity cardinality ($j$-FSSC) for a spectral vector and $j$-fixed band sparsity cardinality ($j$-FBSC) for a spatial band. In contrast to fixed length coding, there is variable length coding that can be used to derive VSC from a 3-D spectral–spatial data cube. Two versions of VSC can also be derived as counterparts of $j$-FSSC and $j$-FBSC. One is $jN$-VSC corresponding to $j$-FSSC which considers spectral–spatial samples in a 3-D data cube with $k$ determined by $j$ and $N$. The other is $jL$-VSC corresponding to $j$-FBSC which considers spatial–spectral band in a 3-D data cube with $k$ determined by $j$ and $L$. Both can be simply represented by $k$-VSC with either $k = j \times N$ or $k = j \times L$ where $j$ is automatically determined by MXSVD. Unlike FSC that fixes $j$ nonzero elements in each spectral vector or each spectral band, VSC allows variable numbers of nonzero elements in a 3-D data cube in both spectral vectors and spectral bands. Using the above interpretations, the SC defined in [29] and [30] turns out to be $jN$-VSC. By coupling SC $= k$ with $j$ ICs produced by ATGP-FastICA, we can indeed resolve the issue between anomalies and noises/interferers to improve the detectability of anomalies in EAS because non-Gaussian noises cannot be sparse and can be removed by either FSC or VSC.

As an illustrative example, RX-AD in (3) can now operate $(\mathbf{r} - \boldsymbol{\mu})^T$ and $\mathbf{K}^{-1}$ on the same data space specified by EAS as follows:

$$\delta^{\text{RX-AD}}(\mathbf{r}_{\text{EAS}}) = (\mathbf{r}_{\text{EAS}} - \boldsymbol{\mu}_{\text{EAS}})^T \mathbf{K}_{\text{EAS}}^{-1} (\mathbf{r}_{\text{EAS}} - \boldsymbol{\mu}_{\text{EAS}}) \quad (4)$$

where there is no need of separating low-rank and sparse spaces in (3) since the low-rank space has been taken care of by ICA.

As noted above, BS and anomaly detectability issues are two key factors in evaluating the effectiveness of an anomaly detector. It turns out that both eventually lead to the BS issue, that is, detection of anomalies in BKG space and detection of anomalies in AS. It has been shown in [5], [9], [32], [33], [47], and [48] that the traditional 2-D receiver operating characteristic (ROC) curve was incapable of evaluating BS because the detection probability $P_D$ and false alarm probability $P_F$ used to plot ROC curves are determined by the same threshold $\tau$. As a consequence, if $P_D$ is high, so

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is $P_F$ and vice versa. In other words, $P_D$ and $P_F$ are tied together simultaneously and cannot work alone independently; thus, they cannot individually measure target detectability (TD) and BKG suppressibility, respectively. This indicates that solely relying on $AUC_{(D,F)}$ is misleading, an issue which was already addressed and demonstrated in [47]. To resolve this dilemma, a 3-D ROC-analysis-based evaluation tool for target detection developed in [47] extends the traditional 2-D ROC analysis by plotting a 3-D ROC curve as a function of three parameters, $P_D$, $P_F$, and $\tau$ where $\tau$ is actually an independent variable and both $P_D$ and $P_F$ are in fact dependent variables on $\tau$. In this case, a 3-D ROC curve is a function of the three parameters, $P_D$, $P_F$, and $\tau$ from which three 2-D ROC curves of ($P_D$,$P_F$), ($P_D$,$\tau$), and ($P_F$,$\tau$) can be further generated along with their respective area under curve (AUC) values, denoted by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$. As a result, $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ can be used to evaluate TD and BKG suppressibility, respectively. In particular, several new measures derived from $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$ in [47] can be used for this purpose.

The main contributions of this article are summarized in the following.

1) It develops an IC component space-based AS, $\mathbf{X}^{AS}_{\mathbf{IC}^j}$, which is formed by the first $j$ ATGP-FastICA produced ICs. This is very different from the traditional ICA-based AD approaches which use individual ICs to perform AD separately.
2) It derives four new versions of SC, two spectral versions of FSC, $j$-FSSC and $j$-FBSC, and two spectral–spatial versions of VSC, *jL*-VSC and *jN*-VSC. In addition, it also designs a toy example to illustrate how to construct EAS via these four versions of SC.
3) Four developed SC versions are further imposed on $\mathbf{X}^{AS}_{\mathbf{IC}^j}$ to construct EAS, $\mathbf{X}^{EAS}_{\mathbf{IC}^j} = \mathbf{X}^{SC}_{\mathbf{IC}^j}$ so that AD performance can be enhanced by operating an anomaly detector in $\mathbf{X}^{EAS}_{\mathbf{IC}^j} = \mathbf{X}^{SC}_{\mathbf{IC}^j}$.
4) It presents experiments to demonstrate that several most recent developed AD techniques can be actually improved by $\mathbf{X}^{EAS}_{\mathbf{IC}^j} = \mathbf{X}^{SC}_{\mathbf{IC}^j}$.
5) A particularly interesting issue is also investigated by experiments to demonstrate an issue of the commonly used dataset, San Diego airport data scene, which is inappropriate to be used for AD.

The remainder of this article is organized as follows. Section II reviews ICA-related works. Section III derives the ATGP-FastICA algorithm along with an automatic method to determine the number of ICs to be retained, $j$, to form $\mathbf{X}^{AS}_{\mathbf{IC}^j}$. Section IV presents a toy example to illustrate the concept of SC and further designs four versions of SC to construct EAS. Section V derives 3-D ROC curve-based detection measures. Section VI describes the images to be used for experiments and presents the experimental results and discussions. Section VII conducts experiments to demonstrate the advantage of using EAS to improve the performance of the current recently developed AD approaches. Section VIII investigates an issue of using San Diego airport data scene for AD. Section IX draws conclusions.

## II. Related Works

Due to their unknown nature anomalies are considered as crucial targets to image analysts and their detection presents a great challenge, particularly in military applications. This is because anomalies are generally embedded in BKG and also corrupted by noise as being sandwiched between BKG and noise. So, for an anomaly detector to work well, it must address both BKG and noise issues effectively [5]. Since RX-AD in (1) and R-AD in (2) use the global covariance/correlation matrix to suppress BKG, it cannot deal with BKG locally. To resolve this issue, RX-AD and R-AD have been extended to the model specified by (3) in [9], [32], [33], and [48].

Unfortunately, all the aforementioned approaches only focused on separation of BKG from anomalies and did not address the follow-up AD. The work in [32] was believed to be first to consider an anomaly detector to be performed in two different data spaces by suppressing BKG in a low-rank space and extracting anomalies in sparse spaces separately. More recently, similar approaches were also extended to orthogonal-subspace-projection-based anomaly detector (OSP-AD) [48] and CEM-AD in [9].

Deviating from the LRaSR model-based approaches, Chen *et al.* developed CDA in [33] which extended RPCA [28] and robust subspace learning (RSL) via S+LR formulation [49] by replacing their sparse spaces with an IC component space to enhance anomaly detectability. This suggests that using the IC component space to detect anomalies may be more effective than using the sparse space. With this in mind, ICA has also been used for AD. Tiwari *et al.* [50] used FastICA to generate all ICs and then used histogram thresholding to segment anomalies from BKG in each of ICs. Jablonski *et al.* [51] developed a global iterative principal component analysis (PCA) reconstruction-error-based anomaly detector to detect anomalies by computing errors (residuals) associated with reconstructing the original image by PCs. Johnson *et al.* [52] developed ICA-based autonomous global anomaly detector (AutoGAD) method which sphered data prior to FastICA and then used feature selection to potential anomalies. Song *et al.* [53] developed a kernel independent component analysis (KICA) to map hyperspectral data into the kernel space for feature extraction and then used FastICA to extract the core IC for RX-AD to detect anomalies. Song *et al.* [54] developed an orthogonalized ICA to generate ICs from which potential anomaly signal-to-noise ratio (PASNR) for each generated IC can be calculated for thresholding to find abnormal information for AD and then used orthogonal subspace projection (OSP) to suppress BKG to enhance AD. All these ICA-based AD approaches made attempts to address the three issues noted in the introduction one way or another, but not all of them together. This is mainly because one major problem with previously developed ICA-based approaches to perform AD is their use of the ICA-generated ICs as separate independent unmixed abundance maps from which features can be extracted for AD in

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each of ICs individually. In this case, each data sample is a single pixel but not a pixel vector in an IC component space as we discuss in this article as a 3-D data cube.

One primary reason that a classic anomaly detector such as RX-AD does not work well is attributed to its inability in dealing with the BKG issue and its detectability of anomalies in the original data space. To cope with these two issues, recent approaches developed for AD have been focused on how to represent BKG and anomalies by various models in terms of LRaSR. Specifically, Candes *et al.* [28] proved that RPCA could be used as a blind source separation technique to decompose the data matrix $\mathbf{X}$ into

$$\mathbf{X} = \mathbf{L} + \mathbf{S} \tag{5}$$

in a matrix decomposition. Using a similar approach to RPCA, Vaswani *et al.* [49] developed an RSL, which treated an outlier as an additive sparse corruption via sparse+low-rank (S+LR) matrix decomposition. With this interpretation, RPCA can be considered as a variant of RSL via S+LR formulation. Another attractive model is LRaSMD that models a data matrix $\mathbf{X}$ as a linear decomposition of a low-rank matrix, **L,** and a sparse matrix, **S,** plus a noise matrix $\mathbf{N}$ represented by

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \tag{6}$$

where GoDec is used to find the low-rank matrix $\mathbf{L}$ and the sparse matrix $\mathbf{S}$. Most recently, CDA [33] represents the data matrix $\mathbf{X}$ as

$$\mathbf{X} = \mathbf{PC} + \mathbf{IC} + \mathbf{N} \tag{7}$$

where **PC** and **IC** are the PC space and IC space, respectively. By taking advantage of models (5–7), anomaly detectors can operate the low-rank **L/PC** space and the sparse **S/IC** space separately to address both BKG and AD issues.

## III. Independent Component Analysis-Based AD

ICA has a wide range of applications in communications/signal processing [34] and video processing [55], [56]. It is a blind source separation technique that can separate blindly mixed signal sources. Its exploration in hyperspectral data has also been studied [57]. Most notable applications are data dimensionality reduction (DR) [58], endmember extraction [59], and spectral unmixing [60]. Specifically, when it comes to AD application, an anomaly can be considered as a rare target. From a statistical point of view, it can be better described by HOS. Since ICA is an HOS-based technique which has removed 2OS, ICA is particularly suitable for AD.

The key idea of ICA assumes that data samples are linearly mixed by a set of separate independent sources so that ICA can unmix these signal sources according to their statistical independency measured by mutual information. To validate its approach, an underlying assumption is that at most one source in the mixture model can be allowed to be a Gaussian source. This is due to the fact that a linear mixture of Gaussian sources is still a Gaussian source. More precisely, let $\mathbf{r}$ be an $L$-dimensional mixed signal source vector expressed by

$$\mathbf{r} = \mathbf{As} \tag{8}$$

where $\mathbf{A}$ is an $L \times p$ mixing matrix, known as abundance fraction matrix, and $\mathbf{s} = (s_1, s_2, \ldots, s_p)^T$ is a $p$-dimensional signal source vector with $p$ signal sources, $\{s_t\}_{t=1}^{p}$, needed to be separated, known as endmembers. The purpose of ICA is to find an unmixing matrix $\mathbf{W}$ that separates the mixed signal source vector $\mathbf{r}$ into a set of $p$ signal sources $\{s_t\}_{t=1}^{p}$ which are statistically independent. Several different criteria have been proposed to measure source independency [34]. Nevertheless, they all originated from the concept of mutual information which is a criterion to measure the discrepancy between two random sources [60].

### A. ATGP-FastICA

Compared with PCA which uses eigenvectors to generate PCs, ICA does not have a corresponding mechanism to generate ICs. So, to find ICs, an algorithm, called FastICA, developed in [34]–[37] has been widely used for this purpose, but it suffers from three major issues summarized as follows.

First, FastICA uses a random initial projection vector to generate each of ICs. As a consequence, its generated ICs are not repeatable. Most importantly, ICs generated earlier by FastICA do not necessarily indicate that they are more significant than those ICs generated subsequently. A second issue is that although ICA can remove Gaussian noises, it does not distinguish non-Gaussian noises/interferers from targets of interest in ICs. So, how to extract targets not noises from ICs is crucial. A third issue is because not all ICs are equally important, how to identify desired ICs for AD is challenging. Apparently, these three issues are particularly critical in designing and developing effective ICA-based anomaly detectors. Unfortunately, despite that many ICA approaches developed for AD have been reported in the literature, not all these issues were addressed.

Due to its use of an iterative process, FastICA requires an initial condition for initialization. Since there are no eigenvalues that can be used to generate an IC vector, it must rely on an initial projection vector to find a final desired projection vector in an iterative manner to represent each of ICs. Two consequences result from this process. One is that different initial conditions produce different IC projection vectors, thus different ICs. The other is that each generated IC has no preference to other ICs. This is quite different from PCs whose significance can be weighted by their corresponding eigenvalues. To address this issue, Wang and Chang [39] used ATGP developed in [38] to generate a set of specific initial projection vectors that can be used to generate each of ICs by FastICA. So, the resulting FastICA, referred to as ATGP-FastICA not only can prioritize all the ICs by the ATGP-generated initial projection vectors but also can resolve the nonrepeatable IC issue caused by random initial conditions.

As also noted in the introduction, prior to implementing ICA, DS [61] is a required preprocessing technique which removes 2OS including Gaussian noise. However, it also comes with a major issue that the ICA-generated ICs do not separate targets of interest from non-Gaussian noises/interferers. Also, because ICs are generated randomly

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according to their use of random initial projection vectors, there are no priority orders assigned to the generated ICs in the same way that PCs are ranked by their corresponding eigenvalues. Accordingly, some ICs that contain non-Gaussian noises/interferers may appear earlier and could be mistakenly treated as targets of interest. Fortunately, ATGP-FastICA can rank all the ICs according to the orders of ATGP-generated initial projection vectors, in which case those ICs containing non-Gaussian noise may be ranked with lower priorities. This is because ICs using ATGP-generated targets as projection vectors will contain targets of interest with their significance ranked based on prioritized ATGP-generates targets. As a result, ICs that contain non-Gaussian noises/interferers are very likely to be those with least priorities ranked by ATGP. If an appropriate number of ICs with higher priorities are selected, then the chance of selecting high-prioritized ATGP-generated ICs containing non-Gaussian noises/interferers would be very low.

### B. Determination of Number of ICs

Unlike many LRaSMD-based methods which require prior knowledge about the ranks of both low-rank and sparse spaces, ATGP-Fast-ICA only requires knowing the number of ICs, $j$, needed to be used to form an IC component space. This is because the BKG specified by the low-rank space has been removed by DS. So, a critical issue is how to determine an appropriate value of $j$.

To take care of this issue, the concept of VD developed in [8] and [41]–[44] can be jointly implemented with an algorithm, called MXSVD developed by Kuybeda *et al.* [45], to determine a desired number of ICs, $j$. The value of $j$ is then used to form a $j$-IC component space as AS, $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ which is supposed to exclude those ICs that contain non-Gaussian noises/interferers because of their priorities lower than $j$. To the author's best knowledge, the ICA-based AD approaches reported in the literature treated ICs as independent and separate components to perform AD, but not use joint $j$ ICs as a data cube, $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, to perform AD as proposed in this article. A significant advantage of working on $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ is to allow an anomaly detector to correlate data samples among different ICs to improve anomaly detectability, a task which cannot be accomplished by working on each individual ICs independently and separately.

Based on the fact that anomalies can be considered as rare signal sources, an approach developed by Kuybeda *et al.* [45] which was particularly designed to preserve anomaly (rare) vectors embedded in a high dimensional noisy data space, can be used for this purpose to estimate the rank of rare signal subspace, that is, $j$. In essence, Kuybeda *et al.*'s approach consists of two major algorithms, called the maximum orthogonal complement algorithm (MOCA), which determines a rank of signal subspace, $p$, and MXSVD, which determines the rank of a rare signal subspace within the signal subspace.

The idea of MOCA assumes that the data space $\mathbf{X}$ is represented by an additive sum of two major components, signal subspace, $\mathbf{Y}$, and noise subspace, $\mathbf{Z}$, as $\mathbf{X} = \mathbf{Y} + \mathbf{Z}$ where the rank of $\mathbf{Y}$ is assumed to be $p$. In addition, the signal subspace, $\mathbf{Y}$, is further assumed to be a decomposition of two subspaces, $\Psi$ and $\Omega$, as $\mathbf{Y} = [\Psi|\Omega]$, with $\Psi$ as the BKG subspace of rank $m$, and $\Omega$ as the rare signal subspace of rank $j$ so that $p = m + j$ where, can be estimated by MXSVD. So, in terms of models specified by (1–3), $\Psi$ and $\Omega$ can be interpreted as the low-rank space $\mathbf{L}$ and the sparse space $\mathbf{S}$, respectively.

However, MOCA uses the maximum likelihood detector to determine the value of $p$ with an assumption that two hypotheses representing signal and noise are equally likely. Apparently, this is not true in practical applications, specifically, hyperspectral imaging. It has been also shown in [43] and [44] that it did not perform well compared with the Neyman–Pearson detector used in estimating VD. So, in this case, VD is used to replace MOCA to determine the value of $p$. In other words, the rank of BKG subspace + rare signal subspace (BKG+RSS), denoted by $\mathrm{rank}^{\mathrm{VD}}_{\mathrm{BKG+RSS}}$, can be estimated by VD as

$$\mathrm{rank}^{\mathrm{VD}}_{\mathrm{BKG+RSS}} = p. \tag{9}$$

Once the value of $p$ in (11) is determined, MXSVD is then used to determine the value of $j$.

## IV. Construction of Effective Anomaly Space by Sparsity Cardinality

The sphered space $\mathbf{X}_{\mathrm{DS}}$ only removes 2OS-characterized BKG. In this case, $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ produced from $\mathbf{X}_{\mathrm{DS}}$ by the ATGP-FastICA-generated $j$-IC component space, $\mathbf{X}_{\mathbf{IC}^j}$, may still contain non-Gaussian noises/interferers because ICA cannot remove HOS non-Gaussian noises/interferers. To resolve this issue, the concept of SC introduced by GoDec in [29] provides a clue of how to create an SC-constrained AS, called effective anomaly space (EAS), denoted by $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j} = \mathbf{X}^{\mathrm{SC}}_{\mathbf{IC}^j}$, which can remove the non-Gaussian noises/interferers from $\mathbf{X}_{\mathbf{IC}^j}$ via SC.

The idea of SC originated from video sequences in [29] by retaining the first $k$ largest nonzero data sample values in the video frames, while setting the rest of the data sample values in the video sequence to zeros. Inspired by SC, we can consider the 3-D $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ as a video sequence with the spectral bands corresponding to video frames. Because noises are random and generally ubiquitous over video frames, we can also expect that the noise has the same ubiquitous property over $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ with very small magnitudes. This implies that the noises cannot be sparse. Based on this fact, combining ATGP-FastICA with VD/MXSVD-estimated $j$ gives rise to a non-Gaussian noise-removed EAS, $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ which not only removes 2OS characterized BKG from $\mathbf{X}$ via DS but also removes HOS non-Gaussian noises/interferers from $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ via SC. So, $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ constructed by ATGP-FastICA and SC is particularly designed to address three issues altogether, viz. prioritizing ICs, determining the value of $j$, and removing non-Gaussian noises/interferers. Most importantly, an anomaly detector operating on $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ in (4) can significantly improve its performance.

### A. How to Construct EAS Via SC

Assume that $N_r$, $N_c$, and $L$ are the number of rows, number of columns, and number of spectral bands,

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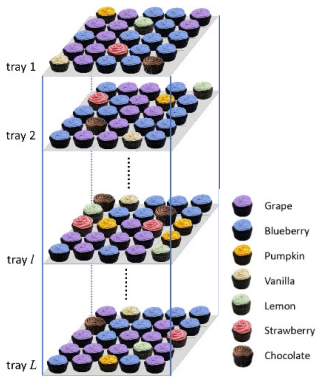



Fig. 1. 3-D rack, R, made up of $N \times L$ muffins with seven different flavors.

respectively. Let $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)} = (r_{(n_r,n_c),1}, r_{(n_r,n_c),2}, \ldots, r_{(n_r,n_c),L})^T$ be a data sample in $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, $\{\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)} = (r_{(n_r,n_c),1}, r_{(n_r,n_c),2}, \ldots, r_{(n_r,n_c),L})^T\}^{N_r,N_c}_{n_r=1,n_c=1}$, where $r_{(n_r,n_c),l}$ is the $l^{\mathrm{th}}$ band value of the data sample $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}$ at spatial location $(n_r, n_c)$. So, there are a total of $N_r \times N_c = N$ data samples in $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, each of which is an $L$-dimensional data sample vector, $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)} = (r_{(n_r,n_c),1}, r_{(n_r,n_c),2}, \ldots, r_{(n_r,n_c),L})^T$.

Since it is difficult to comprehend the utility of SC in [29], we custom-design a muffin (i.e., cupcakes) toy example to illustrate the idea of SC $= k$.

Consider a 3-D rack, R, which has $L$ layers. Each layer in rack R has a tray with $N = N_r \times N_c$ slots to accommodate muffins, one muffin for each slot. So, there are a total of $N \times L$ muffins in the 3-D rack R. Assume that all these $N \times L$ muffins are specified by seven flavors, grape, blueberry, pumpkin, vanilla, lemon, strawberry, and chocolate and ordered by their weights in descending order. Without loss of generality, we assume that all the muffins can be differentiated by their weights, and none of them have the same weight. We then randomly distribute these $N \times L$ muffins to fill up all $N \times L$ slots in rack R as shown in Fig. 1.

Now with rack R in Fig. 1, we can illustrate the concept of SC $= k$ imposed on $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, where each pixel (not pixel vector) in $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ corresponds to one muffin and each band is specified by a tray with size of $N$ slots at a particular layer. Accordingly, there is a one-to-one correspondence between pixels in $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ and muffins in rack R.

According to [29], when SC $= k$ is imposed on $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, it means that the constructed EAS, $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j} = \mathbf{X}^{\mathrm{SC}}_{\mathbf{IC}^j}$, only retains the first $k$ largest data sample values (i.e., pixel values) of $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, while setting the rest of data sample values to zeros. Correspondingly, it is also equivalent to saying that rack R keeps $k$ muffins with the first $k$ largest weights in rack R, and in the meantime, it removes all the remaining muffins from their slots to make their slots empty. The resultant rack is denoted by $\mathrm{R}_{\mathrm{SC}}$, which retains only $k$ muffins in their original spatial locations, $(n_r, n_c)$, while all other slots are empty with no muffins at all. As a result, there is also one-to-one correspondence between $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ and $\mathrm{R}_{\mathrm{SC}}$.

There are two types of SC which can be derived from Fig. 1 via fixed length coding and variable length coding in information theory [46] where fixed length coding yields FSC for a spectral vector, referred to as fixed spectral SC (FSSC), and FSC for a spectral band, referred to as fixed band SC (FBSC), and variable length coding gives rise to VSC for spectral–spatial sample VSC and spatial–spectral band VSC.

*1) j-Fixed Spectral Sparsity Cardinality:* A natural and intuitive approach to constructing an $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ from $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ can be viewed as one from sparse representation. That is, it represents each data sample vector $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}$ by retaining its $j$ largest spectral component values, while setting the remaining spectral component values to zeros. The resulting space is denoted by $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$. More specifically, let $\Omega^j_{\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}}$ be the set of $j$ largest spectral values of $r^j_{(n_r,n_c)}$ in each spectral vector, $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}$, then we can define

$$r^{j-\mathrm{FSSC}}_{(n_r,n_c),l} = \begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l} \in \Omega^j_{\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}} \\ 0, & \text{if } r_{(n_r,n_c),l} \notin \Omega^j_{\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}}. \end{cases} \tag{10}$$

In this case, we can form $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$ according to (15) as

$$\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j} = \left\{ \mathbf{r}^{j-\mathrm{FSSC}}_{(n_r,n_c)} = \left( r^{j-\mathrm{FSSC}}_{(n_r,n_c),1}, r^{j-\mathrm{FSSC}}_{(n_r,n_c),2}, \ldots, r^{j-\mathrm{FSSC}}_{(n_r,n_c),L} \right)^T \right\}^{N_r,N_c}_{n_r=1,n_c=1} \tag{11}$$

which consists of $j$-sparse representations, $\{\mathbf{r}^{j-\mathrm{FSSC}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r=1,n_c=1}$ of $\{\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r=1,n_c=1}$, in $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, and each $\mathbf{r}^{j-\mathrm{FSC}}_{(n_r,n_c)}$ has exactly $j$ nonzero spectral component values. As a result, the resulting sparse space $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$ constructed by $\{\mathbf{r}^{j-\mathrm{FSSC}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r=1,n_c=1}$ is an $L$-dimensional sparse space with $j$ being a sparsity of order. The representation (11) can be considered as an alternative sparse representation to the commonly used sparse representation of $\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}$ constrained by sparsity order $j$, i.e., $||\mathbf{r}^{\mathrm{IC}}_{(n_r,n_c)}||_{l_0} \leq j$ where $||\mathbf{x}||_{l_0}$ is the $l_0$ norm defined as the number of nonzero elements to represent the data sample vector $\mathbf{x}$. So, the number of nonzero pixel values in $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$ is $n_{j\text{-FSSC}} = k$ where it is generally determined by $k = j \times N$ [30].

In what follows, we will show how we use (15) to interpret $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$ using Fig. 1 where there are $N = N_r \times N_c$ pixel vectors as $N$ columns in rack R, and each pixel vector with $L$ components is a column vector in the rack with $L$ trays. In other words, for each pixel vector at the spatial location at $(n_r, n_c)$ in $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$, we have a corresponding column of rack R in Fig. 1 with the pixel $r_{(n_r,n_c),l}$ specified by the muffin in the slot at the spatial location at $(n_r, n_c)$ in the $l$th tray corresponding to band $l$, that is, $\mathrm{muffin}_{(n_r,n_c),l} = r_{(n_r,n_c),l}$ at the $l$th tray. When $j$-FSSC is imposed on $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ according to (15), $\mathbf{X}^{j-\mathrm{FSSC}}_{\mathbf{IC}^j}$ contains only the first $j$ largest values of $r_{(n_r,n_c),l}$ in each pixel vector, $\mathbf{r}^{j-\mathrm{FSSC}}_{(n_r,n_c)}$ of $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$, with all other elements set to zeros. Correspondingly, each column of rack R only keeps muffins with the first $j$ largest weights, while removing all the remaining muffins from their slots in the column. Accordingly, the resulting rack, $\mathrm{R}_{\mathrm{SC}}$, contains only $j$ muffins in each column in their specific trays at their original spatial locations, $(n_r, n_c)$, empty in all slots at other trays. So, there is a one-to-one correspondence between $\mathbf{X}_{\mathrm{SC}}$ and $\mathrm{R}_{\mathrm{SC}}$.

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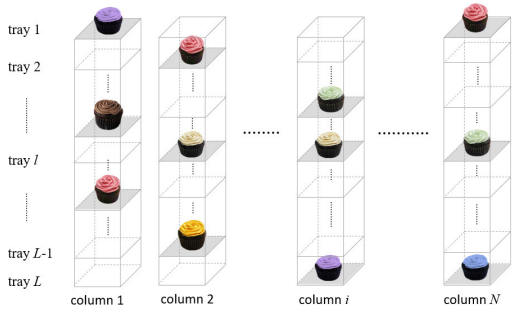



Fig. 2. $j$-fixed spectral SC ($j$-FSSC) with $j = 3$.

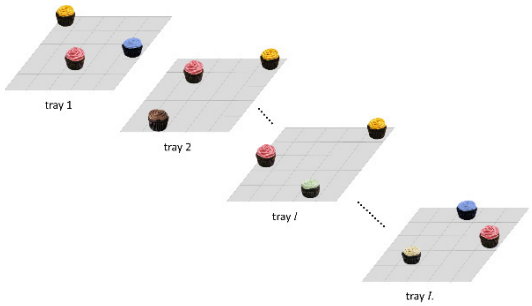



Fig. 3. $j$-fixed band SC ($j$-FBSC) with $j = 3$.

Using Fig. 1, Fig 2 shows how $j$-FSSC works with $j = 3$ and there are only three muffins with the three largest weights among all the muffins in a single column of rack R where each column of the rack corresponds to one pixel vector in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$.

As shown in Fig. 2, there for four columns corresponding to four pixel vectors, each of which has three muffins located at different trays in a single column. Since there are $N$ pixel vector with $L$ spectral components, this indicates that there are also $N$ columns, each of which has $L$ trays. So, with the interpretation of $j$-FSSC, each column corresponding to one single pixel vector has only $j = 3$ muffins retained on their slots in different trays, while muffins in all other slots in the remaining trays have been removed where a blank square represents an empty slot with muffins being removed. In this case, there are only $n_{j\text{-FSSC}} = j \times N$ muffins with $j = 3$.

*2) j-Fixed Spatial Band Sparsity Cardinality:* As an alternative to the $j$-FSSC derived in Section III-A, we can also construct a $j$-fixed spatial band sparsity cardinality ($j$-FBSC) for each band image, $\mathbf{B}_l = [r_{(n_r,n_c),l}]_{n_r=1,n_c=1}^{N_r,N_c}$ in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$. Let $\Omega_l^j$ be the set of $j$ largest pixel values among $\{r_{(n_r,n_c),l}\}_{n_r=1,n_c=1}^{N_r,N_c}$ in the $l$th band image $\mathbf{B}_l$, then we can define $\mathbf{B}_l^{j-\mathrm{FBSC}}$ for the $l$th band image $\mathbf{B}_l$ by

$$r_{(n_r,n_c),l}^{j-\mathrm{FBSC}} = \begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l} \in \Omega_l^j \\ 0, & \text{if } r_{(n_r,n_c),l} \notin \Omega_l^j. \end{cases} \tag{12}$$

In this case, we can construct $\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FBSC}}$ according to (12) as

$$\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FBSC}} = \left\{\mathbf{B}_l^{j-\mathrm{FBSC}}\right\}_{l=1}^{L} \tag{13}$$

which consists of $j$-sparse representations of $\mathbf{B}_l$ in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$ with each $\mathbf{B}_l^{j-\mathrm{FBSC}}$ having exactly $j$ nonzero pixel values. So, unlike $\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FSSC}}$, the number of nonzero pixel values in $\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FBSC}}$ is $n_{j\text{-FBSC}} = j \times L$.

Using Fig. 1 again, Fig. 3 shows $j$-FBSC with $j = 3$ with three muffins with largest weights among all the muffins in a single tray of rack R where each tray of the rack R corresponds to one band image in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$.

As shown in Fig. 3, there for four trays corresponding to four band images, each of which has three muffins located in different slots at the same tray. Since there are $L$ band images with each band image having $N$ spatial locations, this indicates that there are also $L$ trays with each tray having $N$ slots. So, $j$-FBSC can be interpreted as each tray corresponding to one single band image which has only $j = 3$ muffins retained in their slots at their original spatial locations in a single tray, while the muffins in all other slots in the tray at the remaining spatial locations have been removed with a blank square representing an empty slot with muffins being removed from the tray. In this case, there are only $n_{j\text{-FBSC}} = j \times L$ muffins with $j = 3$.

*3) Spectral–Spatial Sample VSC:* The spectral SC, $j$-FSSC, in Section IV-A.1 and $j$-FBSC in Section IV-A.2 impose SC spectrally on each spectral vector and spatially on each band by fixing the value of $j$, respectively. This section derives a quite different version of SC in the sense that SC $= k$ is imposed on the 3-D AS data cube by varying $k$ nonzero elements in either spatial or spectral dimensions. It is called $k$-variable sparsity cardinality ($k$-VSC) which is exactly the same one proposed in [29]. Unfortunately, the value of $k$ was empirically determined in [29]. In [30], this $k$ is further specified by $k = j \times N = jN$ where $j$ is the rank of sparse matrix $\mathbf{S}$ which was also empirically determined. In our derived $k$-VSC, the value of $k$ is specified determined by $k = jN$ with $j$ and $N$ being the number of ICs and the total number of data samples, respectively. Accordingly, the resulting $k$-VSC is actually spectral–spatial sample VSC and can be more specifically defined as $jN$-VSC. Interestingly, such $jN$-VSC can be considered as a variable version of $j$-FSSC.

For a given $k$, we define $\Omega$ as the set consisting of the $k$ largest nonzero pixel values among all $\{r_{(n_r,n_c),l}\}_{n_r=1,n_c=1,l=1}^{N_r,N_c,L}$ in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$. In this case, some of $r_{(n_r,n_c),l}$ in $\Omega$ may belong to the same data sample vector $\mathbf{r}_{(n_r,n_c)}^{\mathrm{IC}}$ in different spectral band components. Accordingly, $k$ can be specified by $k = j \times N$ corresponding to $j$-FSSC. In this case, we can define a $k$-VSC space, $\mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}}$, of $\mathbf{X}_{\mathbf{IC}^j}$ as

$$\mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}} = \left\{\mathbf{r}_{(n_r,n_c)}^{\mathrm{VSC}} = \left(r_{(n_r,n_c),1}^{\mathrm{VSC}}, r_{(n_r,n_c),2}^{\mathrm{VSC}}, \ldots, r_{(n_r,n_c),L}^{\mathrm{VSC}}\right)^T\right\}_{n_r=1,n_c=1}^{N_r,N_c} \tag{14}$$

where $\mathbf{r}_{(n_r,n_c)}^{\mathrm{VSC}}$ is a sparse representation of $\mathbf{r}_{(n_r,n_c)}^{\mathrm{IC}}$ specified by

$$r_{(n_r,n_c),l}^{\mathrm{VSC}} = \begin{cases} r_{(n_r,n_c),l}, & \text{if } r_{(n_r,n_c),l} \in \Omega \\ 0, & \text{if } r_{(n_r,n_c),l} \notin \Omega. \end{cases} \tag{15}$$

Those data sample vectors not in $\mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}}$ but in $\mathbf{X}_{\mathbf{IC}^j}$ are considered as not potential anomalous data samples.

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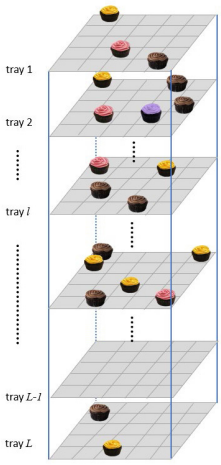



Fig. 4. $3N$-VSC with $k = 3 \times N$.

For any data sample vector in $\mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}}$, $\mathbf{r}_{(n_r,n_c)}^{\mathrm{VSC}}$ is the $k$-VSC representation of $\mathbf{r}_{(n_r,n_c)}^{\mathrm{IC}}$. If we define $|A|$ as the number of elements in set $A$, then $|\{r_{(n_r,n_c),l}\}_{l=1}^{L} \cap \mathbf{\Omega}| = k$. For example, if $\mathbf{r}_{(n_r,n_c)}^{\mathrm{VSC}}$ has $b$ spectral components, i.e., $\|\mathbf{r}_{(n_r,n_c)}^{\mathrm{VSC}}\|_{l_0} = b$ which are nonzero. In this case, $\{r_{(n_r,n_c),l_i}\}_{i=1}^{b} \subset \mathbf{\Omega}$. So, the number of nonzero pixel values in $\mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}}$ is $n_{k\text{-VSC}} = k$.

Now, using Fig. 1 again for illustration, Fig. 4 shows $jN$-VSC with $n_{k\text{-VSC}} = k = j \times N$.

Unlike Fig. 2 which retains three muffins in each single column and Fig. 3 which retains three muffins in each tray, Fig. 4 shows that each tray retains different numbers of muffins and all the retained muffins in the trays are among the $k$ largest weights. For example, there are five different trays representing five different band images, each of which retains different numbers of muffins at different spatial locations. In particular, there are three muffins in tray 1, five muffins in tray 2, four muffins in tray $l$, no muffins in tray $L$-1, and two muffins in tray $L$. Nevertheless, Figs 2 and 4 have the same number of $k$ muffins with $n_{j\text{-FSSC}} = 3 \times N$ in Fig. 2 and $n_{k\text{-VSC}} = k = 3 \times N$ in Fig. 4. However, this number is different from the number of muffins $n_{j\text{-FBSC}} = 3 \times L$ in Fig. 3.

It should be noted that in $k$-VSC, there may be a single column containing less than three muffins such as only two muffins in column $N$ corresponding to the pixel vector $N$ in Fig. 4 or one single tray with no muffins in slots at all such as tray $L$-1 corresponding to band $-1$ in Fig. 4. Unlike $j$-FSSC which has exactly $j$ muffins in each column and $j$-FBSC which has exactly $j$ muffins in each tray, $k$-VSC in Fig. 4 has exactly $k$ muffins distributed over the slots in $L$ trays with various numbers of muffins in different trays. So, both $j$-FSSC and $k$-VSC have exactly a total number of $k = j \times N = jN$ muffins.

*4) Spatial–Spectral Band VSC:* Similar to $jN$-VSC, we can also derive a fourth version of SC, to be called spatial–spectral band VSC, $jL$-VSC with $k$ determined by $k = j \times L = jL$, i.e., the number of spectral bands, $L$. It can be considered as a variable version of $j$-FBSC.

Since the only difference between $jL$-VSC and $jN$-VSC is the value of $k$ determined by either $k = jL$ or $k = jN$, the same example in Fig. 4 can also be used to illustrate the concept of $jL$-VSC by replacing $k = j \times N$ with $k = j \times L = jL$. However, it should also be noted that the same interpretation cannot be applied to $j$-FSSC and $j$-FBSC because $j$-FSSC is applied spectrally to a spectral vector, while $j$-FBSC is applied spatially to a spectral band compared with $jN$-VSC and $jL$-VSC, both of which are applied to the same 3-D AS data cube. Nevertheless, the parameters $N$ of $jN$-VSC and $L$ of $jL$-VSC have quite different impacts on BS of AD as will be shown in experiments in Section VI.

Finally, we would like to point out that the key novelty of this section is to use fixed length coding and variable length coding from information theory to derive four versions of SC, $j$-FSSC, $j$-FBSC, $jL$-VSC, and $jN$-VSC. As expected, variable length coding generally performs better than fixed length coding, which is exactly what experiments will show in Section VI. The same argument can also be applied to LRaSR whose sparse representation is constrained by a fixed value as a fixed length coding. Accordingly, it is anticipated that it may not compete $k$-VSC as also shown by experiments in Section VI.

### B. AD on EAS

The EAS presented above is obtained by imposing SC on $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$ to yield $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}} = \mathbf{X}_{\mathbf{IC}^j}^{\mathrm{SC}}$. By performing AD in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}}$, the performance can be significantly improved and enhanced. The following algorithm, Algorithm 1, summarizes how to implement ATGP-FastICA to construct $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}}$ followed by an anomaly detector, such as RX-AD and CEM-AD to perform AD.

**Algorithm 1** Implementing an Anomaly Detector on Effective Anomaly Space

1. Initial conditions: $j$ is given
2. Apply ATGP-FastICA to $\mathbf{X}_{\mathrm{DS}}$ to find the first $j$ IC vectors $\{\mathbf{u}_s\}_{s=1}^{j}$ and form $\mathbf{U}_j^{ICA} = [\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_j]$.
3. Find $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}} = P_{\mathbf{U}_j^{ICA}} \mathbf{X}_{\mathrm{DS}}$ where
$$P_{\mathbf{U}_j^{\mathrm{ICA}}} = \mathbf{U}_j^{\mathrm{ICA}} ((\mathbf{U}_j^{\mathrm{ICA}})^T \mathbf{U}_j^{\mathrm{ICA}})^{-1} (\mathbf{U}_j^{\mathrm{ICA}})^T. \quad (16)$$
4. Determine $k$ for SC.
   Let the sparsity cardinality $k$ be specified by either $k = j \times N$ for $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}} = \mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FSSC}} / \mathbf{X}_{\mathbf{IC}^j}^{k-\mathrm{VSC}}$ or $k = j \times L$ for $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}} = \mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FBSC}}$.
5. Let $\Omega$ be a set of the first $k$ largest non-zero pixel values of the 3D data cube, $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$, and then construct an SC-constrained IC sparse space of $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$, denoted by $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}}$, which is made up of the first $k$ largest nonzero pixel samples in $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$ in $\Omega$.
6. Implement an anomaly detector on $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}}$.

## V. 3-D ROC Curve-Derived Detection Measures

The 2-D ROC curve has been widely used to measure performance analysis of signal detection [61]. It is a plot of detection probability $\mathrm{P_D}$ versus false alarm probability $\mathrm{P_F}$. Its AUC, denoted by $\mathrm{AUC_{(D,F)}}$, is calculated as an indicator to

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measure how effective a detector is. Unfortunately, according to a recent work in [47], solely relying on $AUC_{(D,F)}$ is not creditable. This is because both $P_D$ and $P_F$ are calculated by the same threshold $\tau$ used by the detector. As a result, when $P_D$ and $P_F$ are very high, its calculated $AUC_{(D,F)}$ is also very high. Similarly, both $P_D$ and $P_F$ are very low, its calculated $AUC_{(D,F)}$ is also very low. In other words, $P_D$ and $P_F$ are tied together simultaneously and cannot work independently. Consequently, $P_D$ and $P_F$ cannot individually measure TD and BKG suppressibility (BS), respectively. To address this issue, [47] developed an effective 3-D ROC analysis-based evaluation tool, which extends the traditional 2-D ROC analysis by including the threshold $\tau$ as an additional independent parameter to represent a 3-D ROC curve as a function of three parameters, $P_D$, $P_F$, and $\tau$. As a result, a 3-D ROC curve can be generated by a triplet parameter vector specified by $(P_D,P_F,\tau)$ and it can also be used to three 2-D ROC curves of $(P_D,P_F)$, $(P_D,\tau)$, and $(P_F,\tau)$ with their respective AUC values, denoted by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$, where $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ can be used to evaluate TD and BS, respectively. In addition to these three AUC values, five new AUC measures developed in [5], [32], [33], [47], and [48] are shown to be very effective in measuring joint TD, joint BS, target detection and BKG suppression (TDBS), signal-to-noise probability ratio (SNPR), and overall detection performance (ODP) defined in the following.

1) $AUC_{(D,F)}$: effectiveness of a detector
2) $AUC_{(D,\tau)}$: TD of a detector
3) $AUC_{(F,\tau)}$: BS of a detector
4) $AUC_{TD}$: joint TD of a detector defined by

$$0 \le AUC_{TD} = AUC_{(D,F)} + AUC_{(D,\tau)} \le 2. \quad (17)$$

5) $AUC_{BS}$: joint BS of detector defined by

$$-1 \le AUC_{BS} = AUC_{(D,F)} - AUC_{(F,\tau)} \le 1. \quad (18)$$

6) $AUC_{TDBS}$: TDBS of a detector defined by

$$-1 \le AUC_{TDBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 1. \quad (19)$$

7) $AUC_{SNPR}$: SNPR of a detector defined by

$$0 \le AUC_{SNPR} = \frac{AUC_{(D,\tau)}}{AUC_{(F,\tau)}}. \quad (20)$$

8) $AUC_{ODP}$: ODP of a detector defined by

$$-1 \le AUC_{ODP} = AUC_{(D,F)} + AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 2. \quad (21)$$

To summarize, the above detection measures can be generally categorized into the following three functionalities.

1) TD: $AUC_{(D,\tau)}$, $AUC_{TD}$.
2) BKG suppressibility (BS): $AUC_{(F,\tau)}$, $AUC_{BS}$, $AUC_{SNPR}$.
3) Detector's effectiveness: $AUC_{(D,F)}$, $AUC_{TDBS}$, $AUC_{ODP}$.

For those who are interested in using the 3-D ROC analysis developed in [47] for detection performance evaluation, a software package of calculating the above detection measures is available on the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL) at the University of Maryland, Baltimore County (UMBC), https://wiki.umbc.edu/display/rssipl/10.+Download.

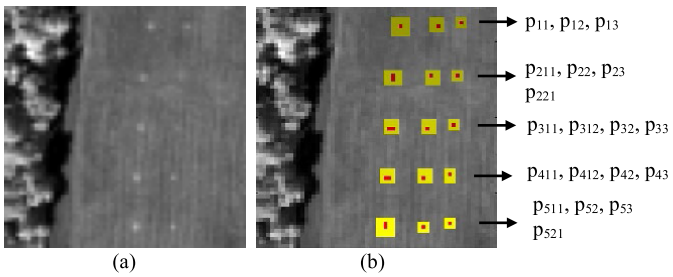



Fig. 5. (a) HYDICE15 panel scene which contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels.

## VI. Hyperspectral Image Experiments

Three hyperspectral real image scenes were used for experiments.

### A. Real Image Scenes Used for Experiments

*1) HYDICE 15-Panel Scene:* An airborne hyperspectral digital imagery collection experiment (HYDICE) scene is shown in Fig. 5. It was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling distance approximately 1.56 m.

This scene has been studied extensively by many reports such as [3], [8], [62]. It has a total of 169 bands and was used for experiments with low-signal/high-noise bands: bands 1–3 and bands 202–210; and water vapor absorption bands: bands 101–112 and bands 137–153 removed. There are 15 square panels in Fig. 5(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively. Due to the ground sampling distance being approximately 1.56 m, each of the panels in the first column except the one in the first row contains two panel pixels highlighted by red, $p_{211}$ and $p_{221}$ in row 2, $p_{311}$ and $p_{312}$ in row 3, $p_{411}$ and $p_{412}$ in row 4, and $p_{511}$ and $p_{521}$ in row 5 as shown in Fig. 5. All the remaining 11 panels in Fig. 5(a) contain single-pixel panels highlighted by red, $p_{11}$, $p_{12}$, and $p_{13}$ in row 1, $p_{22}$ and $p_{23}$ in row 2, $p_{32}$ and $p_{33}$ in row 3, $p_{42}$ and $p_{43}$ in row 4, and $p_{52}$ and $p_{53}$ in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 5(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG.

*2) HYDICE Urban Scene:* A second dataset is another HYDICE image with pseud-color shown in Fig. 6(a). It is an urban scene and comprises 174 bands with the noise and water absorption bands removed from 210 bands. A region with a size of 80 × 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 6(b) along with the ground-truth map shown in Fig. 6(c) where 21 pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from BKG.

*3) San Diego Airport Data:* A third dataset used for experiments was an airborne visible/infrared imaging spectrometer (AVIRIS) data, which is a San Diego airport scene in CA, USA, with pseudo-color shown in Fig. 7(a). It has a size of 400 × 400 pixels with a 3.5-m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the

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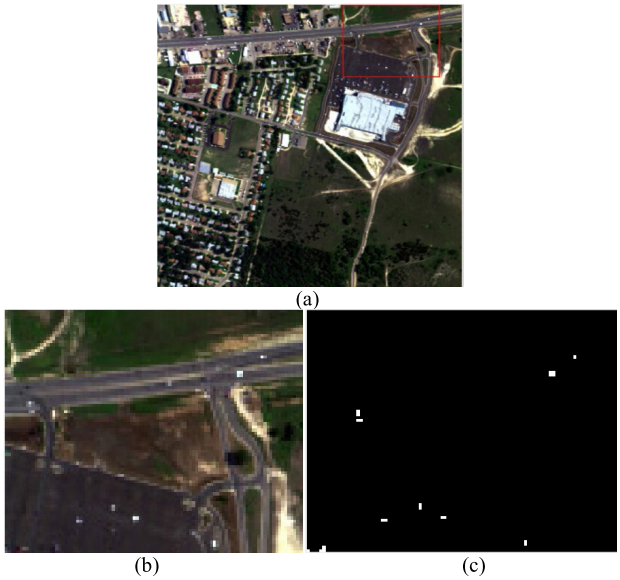



Fig. 6. HYDICE urban scene. (a) Pseudo-color image of the whole scene. (b) Pseudo-color image of the selected area. (c) Ground-truth map.

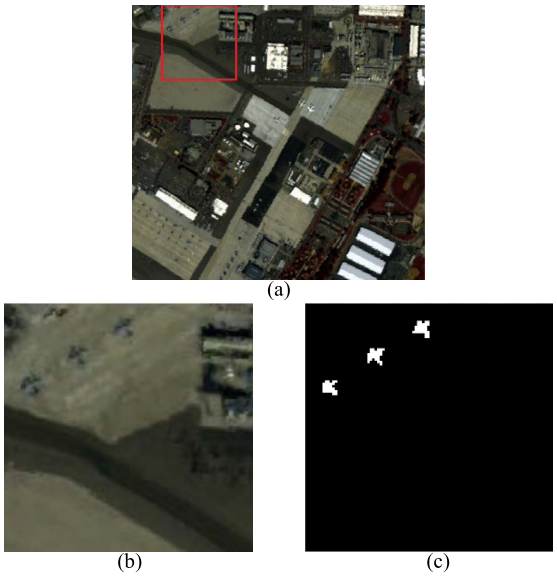



Fig. 7. San Diego Airport scene. (a) Pseudo-color image of the whole scene. (b) Pseudo-color image of the selected area. (c) Ground-truth map.

water absorption regions, low signal-to-noise ratio, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 bands of the data were retained and used for experiments. An area of 100 × 100 pixels at the upper left corner of the scene was selected as the test image shown in Fig. 7(b) along with its ground-truth map shown in Fig. 7(c). It is an urban scene in which the main BKG materials are roof, shadow, and grass. There are three airplanes in the image, which consist of 58 pixels and account for 0.58% of the test image as shown in Fig. 7(b). This scene is less interesting than the HYDICE 15-panel data in Fig. 5 because the only available ground truth is three airplanes which are very large targets. In Section VIII, we will demonstrate by experiments that this scene is not appropriate for AD but rather suitable for target detection.

TABLE I
$p$ AND $m$ ESTIMATED FOR THREE DATASETS

| | $p$ | $j$ by MXSVD |
|---|---|---|
| HYDICE 15-panel scene | 13 by NWHFC | 6 |
| HYDICE urban scene | 11 by HFC | 4 |
| San Diego airport scene | 10 by HFC | 8 |

TABLE II
COMPUTER ENVIRONMENT

| computer environment parameter | configuration |
|---|---|
| OS Name: | Microsoft Windows 10 Home |
| OS Version: | 10.0.19043 N/A Build 19043 |
| OS Manufacturer: | Microsoft Corporation |
| System Type: | x64-based PC |
| Processor(s): | Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz 2.21 GHz |
| Total Physical Memory: | 32,624 MB |
| Graphics Card: | GeForce GTX 1060 Graphics Cards |

*4) Parameters Setting:* Table I shows the value of $p$ estimated by the Harsanyi-Farrand-Chang (HFC) and noise-whiten HFC (NWHFC) methods [8] and [40] with the false alarm probability, $P_F = 10^{-3}$, and the value of $j$ estimated by MXSVD for all the three scenes to be used for GoDec and Table II provides details of the computer environment used for experiments.

### B. Experimental Results and Discussions

To evaluate the effectiveness of $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{EAS}}$, two types of anomaly detectors were used for experiments, RX-AD in (1) [6], CEM-AD in (2) [9], and CDA-AD in (7) [33]. Imposing four versions of SC, $j$-FSSC, $j$-BSSC, $jL$-VSC, and $jN$-VSC, derived in Section IV on $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{AS}}$ results in $\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FSSC}}$, $\mathbf{X}_{\mathbf{IC}^j}^{j-\mathrm{FBSC}}$, $\mathbf{X}_{\mathbf{IC}^j}^{jL-\mathrm{VSC}}$, and $\mathbf{X}_{\mathbf{IC}^j}^{jN-\mathrm{VSC}}$. Implementing RX-AD, CEM-AD, and CDA-AD on these three SC-constrained EASs yields their corresponding nine variants as follows. For simplicity, ATGP-FastICA is abbreviated by IC

1) RX-AD

$$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FSSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{IC-}j\text{-FBSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{IC-}jL\text{-VSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{IC-}jN\text{-VSC}}(\mathbf{r}).$$

2) CEM-AD

$$\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FSSC}}(\mathbf{r}),\ \delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FBSC}}(\mathbf{r}),\ \delta^{\text{CEM-AD}}_{\text{IC-}jL\text{-VSC}}(\mathbf{r}),\ \delta^{\text{CEM-AD}}_{\text{IC-}jN\text{-VSC}}(\mathbf{r}).$$

3) CDA-AD

$$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FSSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FBSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{CDA-}jL\text{-VSC}}(\mathbf{r}),\ \delta^{\text{RX-AD}}_{\text{CDA-}jN\text{-VSC}}(\mathbf{r}).$$

It should be noted that CDA [33] and CEM-AD [9] have been shown to perform at least or better than many other anomaly detectors such as collaborative representation detector (CRD) [16], RPCA [28], LRaSMD-based Mahalanobis distance anomaly detector (LSMAD) [30], orthogonal subspace projection-anomaly detector (OSP-AD) [48], low-rank and

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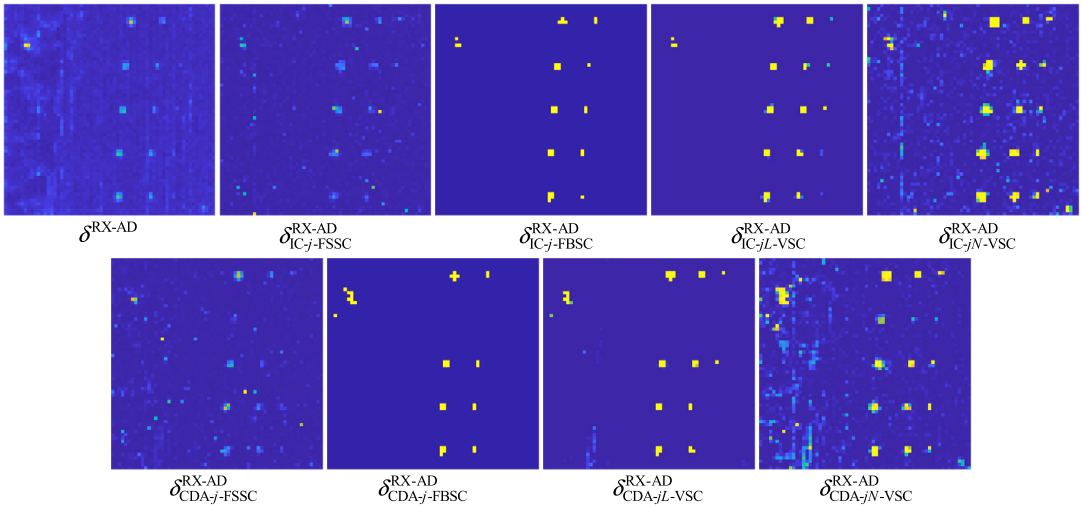



Fig. 8. Detection maps of the HYDICE 15-panel scene using RX-AD and RX-AD on ICSC and CDASC with $p = 13$ and $j = 6$.

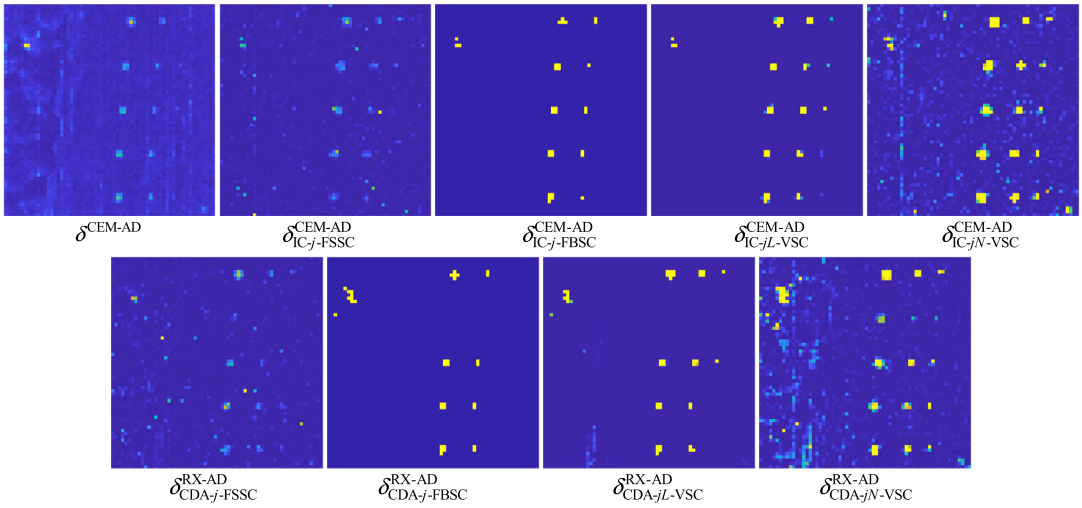



Fig. 9. Detection maps of the HYDICE 15-panel scene using CEM-AD and CEM-AD on ICSC and CDASC with $p = 13$ and $j = 6$.

sparse decomposition model with a mixture of Gaussians (LSDM-MoGs) [25], and prior-based tensor approximation (PTA) [31]. Their experimental results are not included in this article due to limited space. For those who are interested in more detail of comparative analysis and study, we refer to refs. [9], [32], [33].

*1) HYDICE 15-Panel Scene:* Figs. 8 and 9 show the detection maps of the HYDICE 15-panel scene produced by RX-AD and CEM-AD with/without ICSC and CDASC-constrained EAS, $\mathbf{X}^{\text{EAS}}_{\mathbf{IC}^j}$, respectively, with the value of $j$ tabulated by Table I, while Tables III and IV tabulate quantitative results calculated from the eight detection measures derived from 3-D ROC curve analysis presented in Section V where the best results are boldfaced. Apparently, both RX-AD and CEM-AD performed very closely in all the cases. According to Tables III and IV, the best performance was produced by either IC-$k$-VSC with $k = jL$ or IC-$k$-VSC with $k = jN$. However, by visual inspection of Figs. 8 and 9, it seemed that IC-$jL$-VSC did much better than IC-$jN$-VSC in BS but slightly worse than IC-$jN$-VSC in TD, whereas CDA-$jN$-VSC came to the third best anomaly detector with its performance little worse than IC-$jN$-VSC. Interestingly, IC-$j$-FBSC produced the cleanest detection maps with best BS.

*2) HYDICE Urban Scene:* In analogy with the experiments conducted for HYDICE 15-panel data, Figs. 10 and 11 show the detection maps of the HYDICE urban scene produced by RX-AD and CEM-AD with/without ICSC and CDASC-constrained EAS, $\mathbf{X}^{\text{EAS}}_{\mathbf{IC}^j}$, respectively, with the value of $j$ tabulated by Table I, while Tables V and VI tabulate quantitative results calculated from the eight 3-D ROC curve-derived detection measures where the best results are boldfaced. Once again, the similar conclusions drawn from the HYDICE 15-panel data were also applied to the HYDICE urban data. According to Tables III and IV, the best performance was produced by IC-$jL$-VSC with IC-$jN$-VSC as the second best anomaly detector. Also, by visual inspection of Figs. 10 and 11, it is very clear to see that IC-$jL$-VSC did much better than IC-$jN$-VSC in both TD and BS. Although IC-$j$-FBSC produced the best BS, IC-$jL$-VSC produced highest SNPR and much better

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TABLE III

AUC VALUES CALCULATED FROM DETECTION MAPS IN FIG. 8 BY RX-AD USING ICSC AND CDASC FOR FIVE ROWS' DETECTION RESULTS FOR THE HYDICE 15-PANEL DATA SCENE

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{RX-AD}}$ | 0.9898 | 0.3445 | 0.0429 | 1.3342 | 0.9469 | 8.0308 | 0.3016 | 1.2913 | 0.044234 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9587 | 0.2371 | 0.0134 | 1.1958 | 0.9452 | 17.6337 | 0.2237 | 1.1823 | 0.470567 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.8663 | 0.7347 | **0.0094** | 1.6011 | 0.8570 | **78.4065** | 0.7254 | 1.5917 | 0.270093 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jL\text{-VSC}}$ | **0.9956** | 0.8518 | 0.0122 | 1.8475 | **0.9834** | 69.7102 | 0.8396 | 1.8353 | 0.246607 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jN\text{-VSC}}$ | 0.9945 | **0.9932** | 0.0385 | **1.9876** | 0.9560 | 25.7847 | **0.9546** | **1.9491** | 0.217849 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.9817 | 0.2539 | 0.0136 | 1.2356 | 0.9681 | 18.7088 | 0.2404 | 1.2221 | 0.296623 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.7865 | 0.5782 | 0.0108 | 1.3646 | 0.7756 | 53.3896 | 0.5673 | 1.3538 | 0.231253 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.8370 | 0.6826 | 0.0127 | 1.5196 | 0.8243 | 53.6706 | 0.6699 | 1.5069 | **0.215549 s** |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.9932 | 0.8932 | 0.0374 | 1.8863 | 0.9558 | 23.9024 | 0.8558 | 1.8490 | 0.237459 s |

TABLE IV

AUC VALUES CALCULATED FROM DETECTION MAPS IN FIG. 9 BY CEM-AD USING ICSC AND CDASC FOR FIVE ROWS' DETECTION RESULTS FOR HYDICE 15-PANEL DATA SCENE

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{CEM-AD}}$ | 0.9897 | 0.3450 | 0.0434 | 1.3347 | 0.9463 | 7.9485 | 0.3016 | 1.2913 | 0.110456 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9587 | 0.2371 | 0.0135 | 1.1958 | 0.9452 | 17.6144 | 0.2236 | 1.1823 | 0.237596 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.8661 | 0.7345 | **0.0094** | 1.6006 | 0.8568 | **78.3682** | 0.7251 | 1.5912 | 0.218929 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jL\text{-VSC}}$ | **0.9959** | 0.8521 | 0.0122 | 1.8480 | **0.9837** | 69.7387 | 0.8399 | 1.8358 | **0.199504 s** |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jN\text{-VSC}}$ | 0.9945 | **0.9932** | 0.0385 | **1.9876** | 0.9559 | 25.7749 | **0.9546** | **1.9491** | 0.200873 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.9818 | 0.2545 | 0.0136 | 1.2362 | 0.9682 | 18.7543 | 0.2409 | 1.2227 | 0.234484 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.7867 | 0.5784 | 0.0108 | 1.3651 | 0.7759 | 53.4199 | 0.5676 | 1.3543 | 0.215805 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.8370 | 0.6826 | 0.0127 | 1.5196 | 0.8243 | 53.6706 | 0.6699 | 1.5069 | 0.219182 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.9932 | 0.8932 | 0.0374 | 1.8863 | 0.9558 | 23.8851 | 0.8558 | 1.8489 | 0.219918 s |

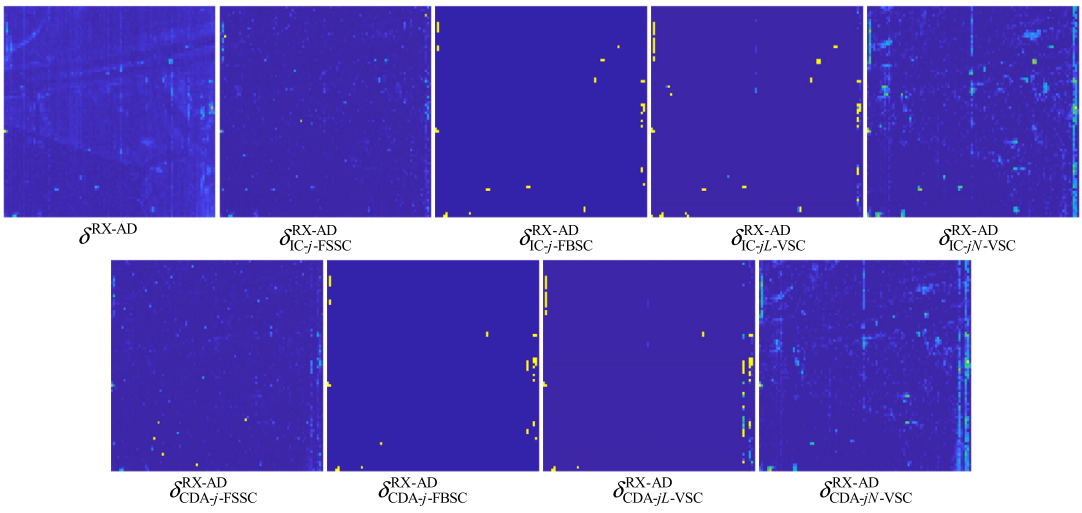



Fig. 10. Detection maps of the HYDICE URBAN scene produced by RX-AD and RX-AD using ICSC and CDASC with $p = 17$ and $j = 6$.

than IC-$j$-FBSC because IC-$jL$-VSC produced much higher TD with very negligible difference in BS compared with IC-$j$-FBSC. The visual assessment was also confirmed by the quantitative results in Tables V and VI.

*3) San Diego Airport Scene:* The San Diego airport scene is quite from the HYDICE datasets considered in Section IX-B.1 and IX-B.2 where the targets of interest are three airplanes. Intuitively speaking, they are not supposed to be anomalies

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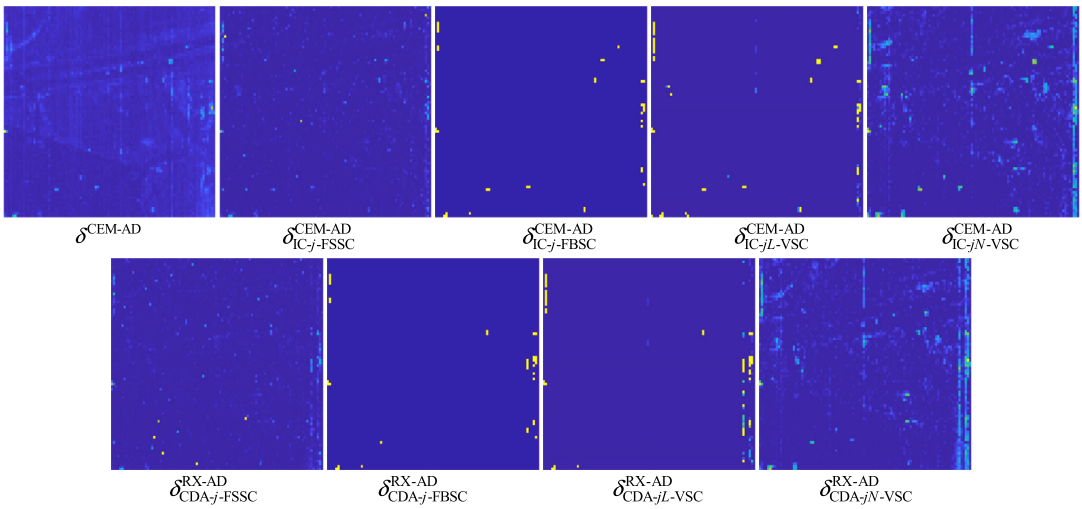



Fig. 11. Detection maps of the HYDICE URBAN scene produced by CEM-AD and CEM-AD using ICSC and CDASC with $p = 17$ and $j = 6$.

TABLE V

AUC VALUES CALCULATED FROM THE DETECTION MAPS IN FIG. 10 BY RX-AD USING ICSC AND CDASC FOR HYDICE URBAN

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{RX-AD}}$ | 0.9864 | 0.2507 | 0.0360 | 1.2371 | 0.9504 | 6.9632 | 0.2147 | 1.2011 | 0.077330 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9563 | 0.1393 | 0.0105 | 1.0956 | 0.9459 | 13.3193 | 0.1288 | 1.0852 | 0.642030 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.7843 | 0.5710 | **0.0080** | 1.3552 | 0.7763 | 71.5675 | 0.5630 | 1.3473 | 0.507995 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jL\text{-VSC}}$ | 0.9258 | **0.8193** | 0.0100 | **1.7451** | 0.9158 | **81.6421** | **0.8093** | **1.7351** | 0.483603 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jN\text{-VSC}}$ | **0.9956** | 0.5721 | 0.0253 | 1.5678 | **0.9703** | 22.5795 | 0.5468 | 1.5424 | 0.403835 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.9441 | 0.0712 | 0.0107 | 1.0153 | 0.9334 | 6.6686 | 0.0605 | 1.0046 | 0.484326 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.5934 | 0.1938 | 0.0088 | 0.7872 | 0.5845 | 21.9084 | 0.1850 | 0.7784 | **0.383307 s** |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.5661 | 0.1464 | 0.0117 | 0.7125 | 0.5544 | 12.5293 | 0.1347 | 0.7009 | 0.388954 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.9799 | 0.3479 | 0.0249 | 1.3277 | 0.9549 | 13.9608 | 0.3229 | 1.3028 | 0.385394 s |

TABLE VI

AUC VALUES CALCULATED FROM THE DETECTION MAPS IN FIG. 11 BY CEM-AD USING ICSC AND CDASC FOR HYDICE URBAN

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{CEM-AD}}$ | 0.9864 | 0.2483 | 0.0359 | 1.2348 | 0.9506 | 6.9238 | 0.2125 | 1.1989 | 0.181546 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9563 | 0.1393 | 0.0105 | 1.0956 | 0.9459 | 13.3209 | 0.1288 | 1.0852 | 0.455172 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.7843 | 0.5710 | **0.0080** | 1.3552 | 0.7763 | 71.5675 | 0.5630 | 1.3473 | 0.386530 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jL\text{-VSC}}$ | 0.9258 | **0.8193** | 0.0100 | **1.7451** | 0.9158 | **81.6421** | **0.8093** | **1.7351** | 0.397629 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jN\text{-VSC}}$ | **0.9956** | 0.5721 | 0.0253 | 1.5678 | **0.9703** | 22.5818 | 0.5468 | 1.5424 | 0.375651 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.9441 | 0.0712 | 0.0107 | 1.0153 | 0.9334 | 6.6694 | 0.0605 | 1.0046 | 0.410308 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.5933 | 0.1936 | 0.0088 | 0.7869 | 0.5844 | 21.8800 | 0.1847 | 0.7780 | 0.426589 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.5661 | 0.1464 | 0.0117 | 0.7125 | 0.5544 | 12.5293 | 0.1347 | 0.7009 | **0.365192 s** |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.9798 | 0.3479 | 0.0249 | 1.3277 | 0.9549 | 13.9559 | 0.3229 | 1.3028 | 0.363461 s |

since these airplanes are so large and very visible. As expected, the conclusions drawn from this data scene would be different from that obtained for the other two HYDICE 15-panel and urban scenes. In particular, RX-AD and

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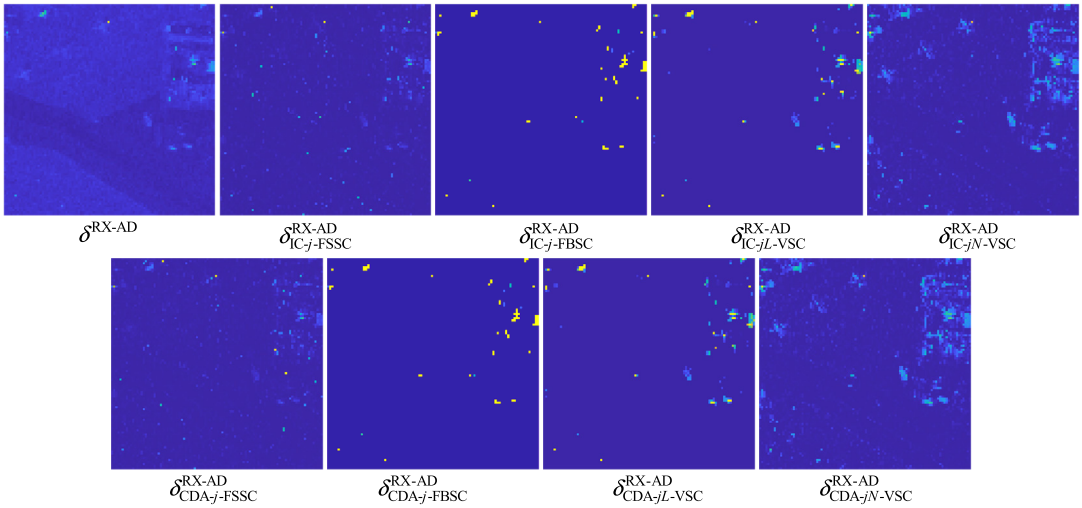



Fig. 12. Detection maps of the San Diego scene produced by RX-AD and RX-AD using ICSC and CDASC.

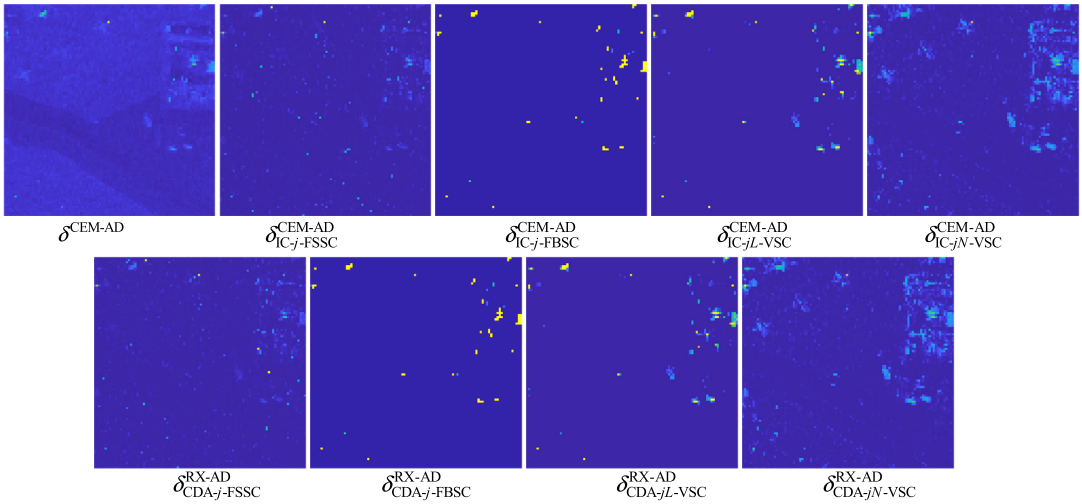



Fig. 13. Detection maps of the San Diego airport scene produced by CEM-AD and CEM-AD using ICSC and CDASC with $p = 10$ and $j = 8$.

CEM-AD performed not as closely as they did for the other two scenes. Figs. 12 and 13 show the detection maps of the San Diego airport scene produced by RX-AD and CEM-AD with/without ICSC and CDASC-constrained EAS, $\mathbf{X}^{\text{EAS}}_{\text{IC}^j}$, respectively, with the value of $j$ tabulated by Table I, while Tables VII and VIII tabulate quantitative results calculated from the eight 3-D ROC curve-derived detection measures where the best results are boldfaced. The only cases which could detect the three airplanes were produced by RX-AD and CEM-AD using IC-*jN*-VSC and CDA-*jN*-VSC. Overall speaking, the best perform was produced by RX-AD and CEM-AD using *jN*-VSC. More detailed studies on this scene will be given in Section VIII.

## VII. Efficacy of EAS in Improving Existing AD

One of most benefits yielded by EAS is to allow an anomaly detector to improve its BKG suppressibility. This section conducted experiments to demonstrate its great potential in enhancement of BKG suppressibility, i.e., anomaly detectability, for several recently develop methods for AD, each of which represents one particular approach and can be reproduced via their provided codes on website.

The first one is the low-rank representation (LRR)-based approach in [27] which optimized BKG dictionary while using regularization parameters to constrain sparse component for anomalies. A second one is the LSDM-MoG used to describe noises in the sparse component [25]. A third one is a PTA [31] which decomposes the data space into two separate spaces, BKG and anomaly tensor spaces. The parameters to implement these methods were exactly the same as that used in their articles. All the experiments were performed according to what exactly they were presented in their articles and then EAS was implemented after their sparse spaces were generated. This practice made sure that these methods were implemented as they were originally designed.

### A. HYDICE Panel Data Scene

Fig. 14 shows the detection maps of the HYDICE panel data scene produced by LSDM-MoG-MD, PTA, and LRR operating

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TABLE VII
AUC VALUES CALCULATED FROM THE DETECTION MAPS IN FIG. 12 BY RX-AD USING ICSC AND CDASC FOR SAN DIEGO AIRPORT

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{RX-AD}}$ | **0.8267** | 0.0905 | 0.0452 | **0.9172** | 0.7815 | 2.0017 | 0.0453 | 0.8720 | 0.095239 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.7119 | 0.0266 | 0.0105 | 0.7384 | 0.7013 | 2.5289 | 0.0161 | 0.7279 | 0.849144 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.5024 | 0.0166 | 0.0114 | 0.5190 | 0.4910 | 1.4618 | 0.0053 | 0.5076 | 0.758877 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jL\text{-VSC}}$ | 0.5144 | 0.0203 | 0.0117 | 0.5347 | 0.5027 | 1.7340 | 0.0086 | 0.5230 | 0.666627 s |
| $\delta^{\text{RX-AD}}_{\text{IC-}jN\text{-VSC}}$ | 0.8151 | **0.0951** | 0.0213 | 0.9102 | **0.7938** | **4.4692** | **0.0738** | **0.8889** | 0.621957 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.6667 | 0.0321 | **0.0104** | 0.6988 | 0.6563 | 3.0682 | 0.0216 | 0.6883 | 0.708130 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.5024 | 0.0166 | 0.0113 | 0.5191 | 0.4911 | 1.4748 | 0.0054 | 0.5078 | **0.580831 s** |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.5084 | 0.0190 | 0.0115 | 0.5274 | 0.4969 | 1.6496 | 0.0075 | 0.5159 | 0.587563 s |
| $\delta^{\text{RX-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.8081 | 0.0851 | 0.0210 | 0.8932 | 0.7871 | 4.0498 | 0.0641 | 0.8722 | 0.608938 s |

TABLE VIII
AUC VALUES CALCULATED FROM THE DETECTION MAPS IN FIG. 13 BY CEM-AD USING ICSC AND CDASC FOR SAN DIEGO AIRPORT

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{CEM-AD}}$ | **0.8274** | 0.0940 | 0.0501 | **0.9214** | 0.7774 | 1.8779 | 0.0439 | 0.8714 | 0.169819 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FSSC}}$ | 0.7119 | 0.0266 | 0.0105 | 0.7385 | 0.7014 | 2.5286 | 0.0161 | 0.7280 | 0.624584 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}j\text{-FBSC}}$ | 0.5024 | 0.0166 | 0.0114 | 0.5190 | 0.4910 | 1.4618 | 0.0053 | 0.5076 | 0.591638 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jL\text{-VSC}}$ | 0.5144 | 0.0203 | 0.0117 | 0.5347 | 0.5027 | 1.7340 | 0.0086 | 0.5230 | 0.561400 s |
| $\delta^{\text{CEM-AD}}_{\text{IC-}jN\text{-VSC}}$ | 0.8151 | **0.0951** | 0.0213 | 0.9102 | **0.7938** | **4.4690** | **0.0738** | **0.8889** | 0.576946 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FSSC}}$ | 0.6659 | 0.0319 | **0.0104** | 0.6979 | 0.6555 | 3.0566 | 0.0215 | 0.6874 | 0.603725 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}j\text{-FBSC}}$ | 0.5024 | 0.0166 | 0.0113 | 0.5191 | 0.4911 | 1.4748 | 0.0054 | 0.5078 | 0.554843 s |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jL\text{-VSC}}$ | 0.5084 | 0.0190 | 0.0115 | 0.5274 | 0.4969 | 1.6496 | 0.0075 | 0.5159 | **0.543664 s** |
| $\delta^{\text{CEM-AD}}_{\text{CDA-}jN\text{-VSC}}$ | 0.8131 | 0.0852 | 0.0210 | 0.8983 | 0.7920 | 4.0548 | 0.0642 | 0.8773 | 0.553323 s |

on AS and EAS using IC-*jL*-VSC and IC-*jN*-VSC. Apparently, by visual inspection, PTA was far worse than LRR and LSDM-MoG-MD.

Table IX tabulates the AUC values of 3-D ROC curves calculated from the detection maps in Fig. 14 with the best results of each of the approaches boldfaced where $j$-FSSC and $j$-FBSC produced the best TD, $AUC_{(D,\tau)}$, and the best BKG suppressibility, $AUC_{(F,\tau)}$, respectively, whereas *jN*-VSC produced either the best or the second best $AUC_{(D,F)}$. It is very obvious to see that operating on EAS significantly improved LSDM-MoG-MD, PTA, and LRR. Nevertheless, compared with Figs. 8 and 9 and Tables III and IV, LSDM-MoG-MD, PTA, and LRR still could not compete against CDA and IC using SC.

### *B. HYDICE Urban Data Scene*

Fig.15 shows the detection maps of the HYDICE urban data scene produced by LSDM-MoG-MD, PTA, and LRR operating on AS and EAS specified by IC-*jL*-VSC and IC-*jN*-VSC. By visual inspection of Fig. 15, PTA was the worst and LRR was the best with LSDM-MoG-MD in the middle.

Table X tabulates the AUC values of 3D ROC curves calculated from the detection maps in Fig. 15 with the best results of each of the approaches boldfaced. Same conclusions drawn from the HYDICE panel data scene were also applied to the HYDICE urban data scene where $j$-FSSC and $j$-FBSC produced the best TD, $AUC_{(D,\tau)}$, and the best BKG suppressibility, $AUC_{(F,\tau)}$, respectively, whereas *jN*-VSC produced either the best or the second best $AUC_{(D,F)}$. Overall speaking, EAS did improve the AD performance of LSDM-MoG-MD, PTA, and LRR. Compared with Figs. 11 and 12 and Tables V and VI, LSDM-MoG-MD, PTA, and LRR generally could not compete against CDA and IC using SC.

### *C. San Diego Airport Data Scene*

Fig.16 shows the detection maps of the San Diego airport scene produced by LSDM-MoG-MD, PTA, and LRR operating on AS and EAS specified by IC-*jL*-VSC and IC-*jN*-VSC. According to the ground truth in Fig. 7(c), the main targets of interest are three airplanes. Then, by visual inspection of Fig. 16, the best performance was yielded by PTA which

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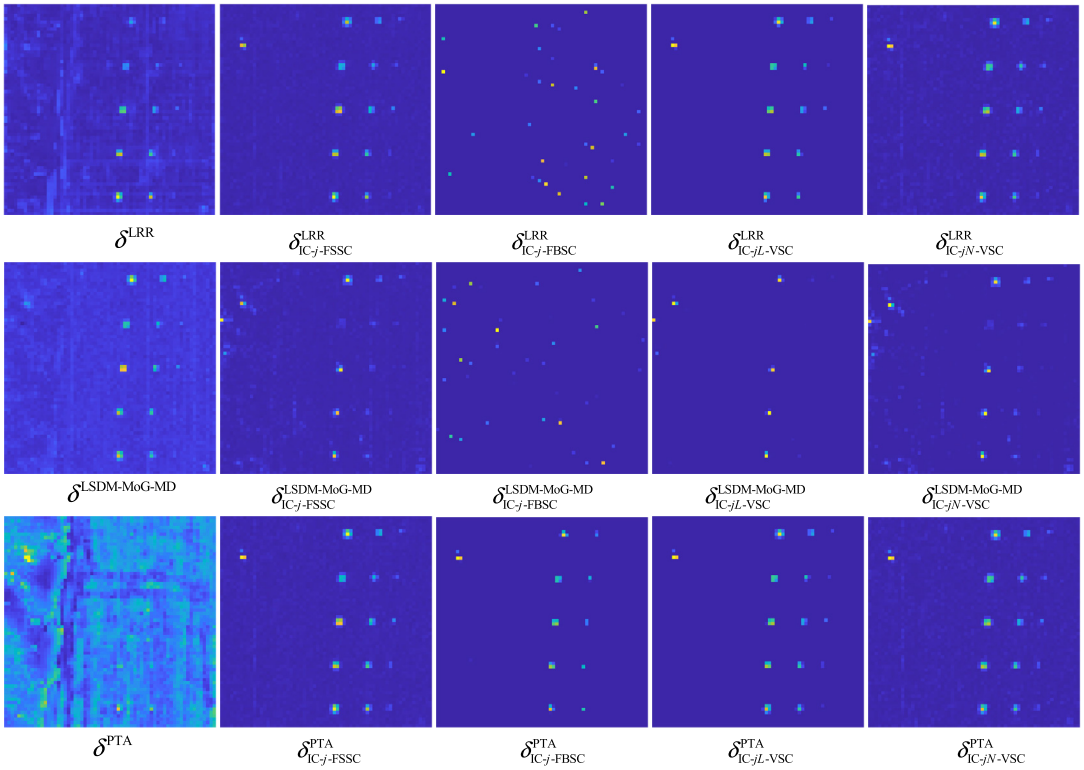



Fig. 14. AD maps produced by LRR, LSDM-MoG-MD, and PTA using various SC criteria for the HYDICE panel data.

TABLE IX
AUC VALUES OF THE HYDICE 15-PANEL SCENE CALCULATED FROM THE DETECTION MAPS IN FIG. 14 BY LRR, LSDM-MOG-MD, AND PTA USING SC CRITERIA

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{LRR}}$ | 0.9882 | 0.4134 | 0.0388 | 1.4016 | 0.9494 | 10.6574 | 0.3746 | 1.3629 | 0.201334 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FSSC}}$ | 0.9972 | 0.5111 | 0.0173 | 1.5083 | 0.9799 | 29.4642 | 0.4937 | 1.4910 | 0.567098 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FBSC}}$ | 0.4955 | 0.0050 | 0.0087 | 0.5005 | 0.4868 | 0.5745 | -0.0037 | 0.4918 | 0.321013 s |
| $\delta^{\text{LRR}}_{\text{IC-}jL\text{-VSC}}$ | 0.9714 | 0.4782 | **0.0074** | 1.4495 | 0.9639 | **64.4446** | 0.4707 | 1.4421 | 0.290028 s |
| $\delta^{\text{LRR}}_{\text{IC-}jN\text{-VSC}}$ | **0.9974** | **0.5203** | 0.0147 | **1.5176** | **0.9827** | 35.4701 | **0.5056** | **1.5030** | **0.278135 s** |
| $\delta^{\text{LSDM-MoG-MD}}$ | **0.9959** | **0.5484** | 0.0658 | **1.5443** | 0.9301 | 8.3360 | **0.4826** | **1.4785** | 0.703802 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9794 | 0.1508 | 0.0087 | 1.1301 | **0.9707** | 17.3762 | 0.1421 | 1.1215 | 0.977964 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FBSC}}$ | 0.4953 | 0.0050 | 0.0073 | 0.5003 | 0.4881 | 0.6864 | -0.0023 | 0.4931 | 1.084519 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jL\text{-VSC}}$ | 0.7083 | 0.1653 | **0.0056** | 0.8736 | 0.7027 | 29.3138 | 0.1596 | 0.8680 | 0.997272 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jN\text{-VSC}}$ | 0.9378 | 0.2100 | 0.0070 | 1.1478 | 0.9307 | **29.8577** | 0.2030 | 1.1407 | **0.938519 s** |
| $\delta^{\text{PTA}}$ | 0.5938 | 0.3703 | 0.2797 | 0.9641 | 0.3141 | 1.3238 | 0.0906 | 0.6844 | 0.155191 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FSSC}}$ | 0.9973 | **0.5395** | 0.0175 | **1.5367** | 0.9797 | 30.7592 | **0.5219** | **1.5192** | 0.357320 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FBSC}}$ | 0.8672 | 0.4082 | **0.0064** | 1.2753 | 0.8608 | **63.8549** | 0.4018 | 1.2689 | 0.337506 s |
| $\delta^{\text{PTA}}_{\text{IC-}jL\text{-VSC}}$ | **0.9974** | 0.4687 | 0.0075 | 1.4661 | **0.9899** | 62.1003 | 0.4611 | 1.4585 | **0.218392 s** |
| $\delta^{\text{PTA}}_{\text{IC-}jN\text{-VSC}}$ | **0.9974** | 0.5029 | 0.0142 | 1.5003 | 0.9832 | 35.3805 | 0.4887 | 1.4861 | 0.219033 s |

was completely opposite to the conclusions made from the HYDICE data scenes in Section VI-A.1 and VI-A.2. This was confirmed by Table XI which tabulates AUC values of 3-D ROC curves calculated from the detection maps in Fig. 16 with the best results of each of the approaches boldfaced. EAS did improve LSDM-MoG-MD, PTA, and LRR in AD performance

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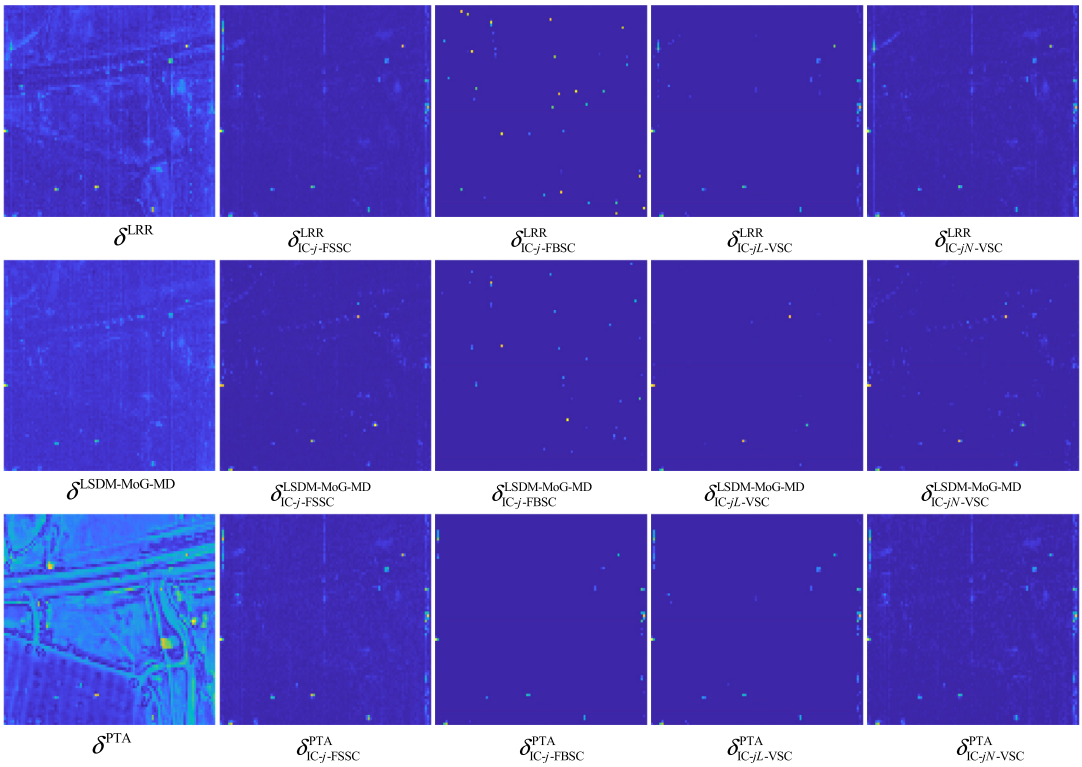



Fig. 15. AD maps produced by LRR, LSDM-MoG, and PTA using various SC criteria for the HYDICE urban data.

TABLE X
AUC VALUES OF HYDICE URBAN CALCULATED FROM THE DETECTION MAPS IN FIG. 15 BY LRR, LSDM-MOG-MD, AND PTA USING SC CRITERIA

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{LRR}}$ | **0.9926** | **0.4864** | 0.0710 | **1.4791** | 0.9217 | 6.8544 | **0.4155** | **1.4081** | 0.137960 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FSSC}}$ | 0.9855 | 0.3264 | 0.0192 | 1.3119 | 0.9663 | 17.0284 | 0.3073 | 1.2927 | 0.552405 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FBSC}}$ | 0.4972 | 0.0050 | 0.0074 | 0.5022 | 0.4899 | 0.6766 | -0.0024 | 0.4949 | 0.433202 s |
| $\delta^{\text{LRR}}_{\text{IC-}jL\text{-VSC}}$ | 0.9026 | 0.2469 | **0.0062** | 1.1495 | 0.8964 | **39.6070** | 0.2407 | 1.1433 | **0.339996 s** |
| $\delta^{\text{LRR}}_{\text{IC-}jN\text{-VSC}}$ | 0.9815 | 0.2945 | 0.0134 | 1.2760 | **0.9680** | 21.9473 | 0.2811 | 1.2626 | 0.368556 s |
| $\delta^{\text{LSDM-MoG-MD}}$ | **0.9788** | **0.3126** | 0.0737 | **1.2914** | 0.9051 | 4.2432 | 0.2389 | **1.2177** | 0.938185 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FSSC}}$ | 0.9234 | 0.2064 | 0.0119 | 1.1299 | 0.9115 | 17.3196 | 0.1945 | 1.1180 | 1.350740 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FBSC}}$ | 0.4975 | 0.0050 | 0.0061 | 0.5025 | 0.4914 | 0.8233 | -0.0011 | 0.4964 | **1.211226 s** |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jL\text{-VSC}}$ | 0.7125 | 0.1540 | **0.0054** | 0.8665 | 0.7071 | 28.4360 | 0.1486 | 0.8611 | 1.314403 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jN\text{-VSC}}$ | 0.9201 | 0.2164 | 0.0070 | 1.1365 | **0.9131** | **31.0423** | **0.2095** | 1.1295 | 1.786017 s |
| $\delta^{\text{PTA}}$ | 0.8008 | **0.4445** | 0.2265 | 1.2453 | 0.5743 | 1.9630 | 0.2181 | 1.0188 | 0.153855 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FSSC}}$ | **0.9966** | 0.4364 | 0.0247 | **1.4330** | 0.9718 | 17.6496 | **0.4117** | **1.4083** | 0.368446 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FBSC}}$ | 0.7604 | 0.1979 | **0.0061** | 0.9583 | 0.7543 | **32.5170** | 0.1918 | 0.9522 | 0.342685 s |
| $\delta^{\text{PTA}}_{\text{IC-}jL\text{-VSC}}$ | 0.9262 | 0.2736 | 0.0065 | 1.1998 | 0.9197 | 41.8647 | 0.2670 | 1.1933 | **0.307539 s** |
| $\delta^{\text{PTA}}_{\text{IC-}jN\text{-VSC}}$ | 0.9955 | 0.3288 | 0.0152 | 1.3243 | **0.9803** | 21.6181 | 0.3136 | 1.3091 | 0.321230 s |

but $AUC_{(D,F)}$ values. Many reports which produced very high $AUC_{(D,F)}$ values for this scene claimed that they performed very well for AD. Unfortunately, this conclusion was misleading as shown in Table XI.

### *D. Discussions on Improvement of EAS on BS*

In Section VII-A and VII-C, experiments were conducted to demonstrate the efficacy of EAS in improving the three most recently developed AD approaches, LRR in [27],

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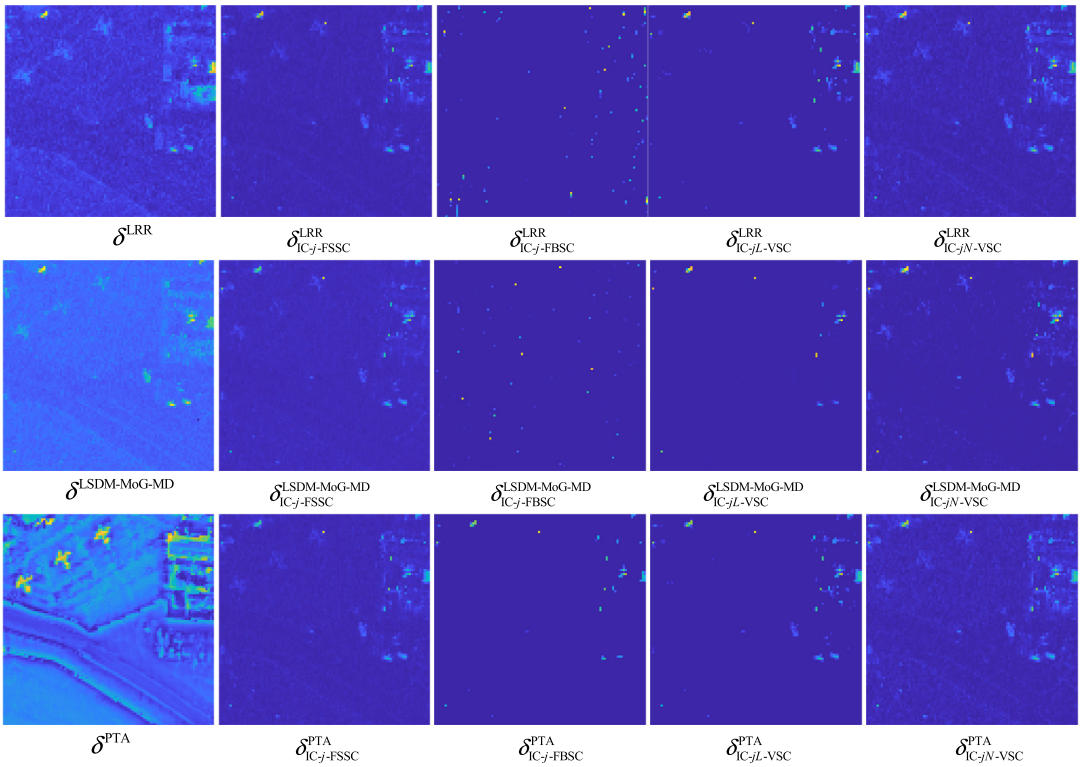



Fig. 16. AD maps produced by LRR, LSDM-MoG-MD, and PTA using various SC criteria for the San Diego airport scene.

TABLE XI
AUC VALUES OF THE SAN DIEGO AIRPORT SCENE CALCULATED FROM THE DETECTION MAPS IN FIG. 16 BY LRR, LSDM-MOG-MD, AND PTA USING SC CRITERIA

| detector | $AUC_{(D,F)}$ | $AUC_{(D,\tau)}$ | $AUC_{(F,\tau)}$ | $AUC_{TD}$ | $AUC_{BS}$ | $AUC_{SNPR}$ | $AUC_{TDBS}$ | $AUC_{ODP}$ | Time |
|---|---|---|---|---|---|---|---|---|---|
| $\delta^{\text{LRR}}$ | 0.8246 | **0.1002** | 0.0642 | 0.9248 | 0.7604 | 1.5612 | 0.0360 | 0.8606 | 0.120422 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FSSC}}$ | **0.8971** | 0.0672 | 0.0252 | 0.9643 | **0.8719** | 2.6654 | 0.0420 | 0.9391 | 0.733642 s |
| $\delta^{\text{LRR}}_{\text{IC-}j\text{-FBSC}}$ | 0.4942 | 0.0050 | **0.0083** | 0.4992 | 0.4859 | 0.6055 | -0.0033 | 0.4909 | 0.656809 s |
| $\delta^{\text{LRR}}_{\text{IC-}jL\text{-VSC}}$ | 0.5373 | 0.0207 | 0.0090 | 0.5580 | 0.5283 | 2.2989 | 0.0117 | 0.5490 | 0.606607 s |
| $\delta^{\text{LRR}}_{\text{IC-}jN\text{-VSC}}$ | 0.8948 | 0.0832 | 0.0275 | **0.9780** | 0.8673 | **3.0225** | **0.0557** | **0.9505** | **0.576663 s** |
| $\delta^{\text{LSDM-MoG-MD}}$ | 0.8605 | **0.2202** | 0.1446 | **1.0807** | 0.7159 | 1.5230 | **0.0756** | **0.9361** | 1.139501 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FSSC}}$ | **0.8679** | 0.0771 | 0.0321 | 0.9450 | **0.8358** | 2.4037 | 0.0450 | 0.9129 | 1.972344 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}j\text{-FBSC}}$ | 0.5006 | 0.0057 | 0.0065 | 0.5063 | 0.4941 | 0.8741 | -0.0008 | 0.4998 | 1.851382 s |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jL\text{-VSC}}$ | 0.5028 | 0.0165 | **0.0063** | 0.5193 | 0.4965 | 2.6412 | 0.0103 | 0.5131 | **1.660603 s** |
| $\delta^{\text{LSDM-MoG-MD}}_{\text{IC-}jN\text{-VSC}}$ | 0.7970 | 0.0518 | 0.0103 | 0.8488 | 0.7866 | **5.0056** | 0.0414 | 0.8384 | 1.671011 s |
| $\delta^{\text{PTA}}$ | **0.9737** | **0.6919** | 0.2358 | **1.6656** | 0.7379 | **2.9346** | **0.4561** | **1.4298** | 0.148972 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FSSC}}$ | 0.8705 | 0.0621 | 0.0275 | 0.9326 | **0.8430** | 2.2603 | 0.0346 | 0.9051 | 0.605682 s |
| $\delta^{\text{PTA}}_{\text{IC-}j\text{-FBSC}}$ | 0.5024 | 0.0167 | **0.0072** | 0.5191 | 0.4953 | 2.3288 | 0.0095 | 0.5120 | 0.555602 s |
| $\delta^{\text{PTA}}_{\text{IC-}jL\text{-VSC}}$ | 0.5186 | 0.0191 | 0.0089 | 0.5376 | 0.5097 | 2.1339 | 0.0101 | 0.5287 | **0.499725 s** |
| $\delta^{\text{PTA}}_{\text{IC-}jN\text{-VSC}}$ | 0.8652 | 0.0751 | 0.0293 | 0.9402 | 0.8359 | 2.5637 | 0.0458 | 0.9110 | 0.536543 s |

LSDM-MoG-MD in [25], and PTA in [31]. This section further discusses their improvement on BS using EAS.

Due to the fact that the traditional 2-D ROC curve of $(P_D,P_F)$ is designed to evaluate the effectiveness of a detector via using the same threshold $\tau$, a higher AUC value of $(P_D,P_F)$ can also be produced by higher values of $P_D$ and $P_F$. As a result, its AUC value of $(P_D,P_F)$ cannot be used to measure the effectiveness of BS. To address this BS issue, a 3-D ROC curve developed in [47] was generated from Figs. 14 to 16 where three 2-D ROC curves were also generated for performance

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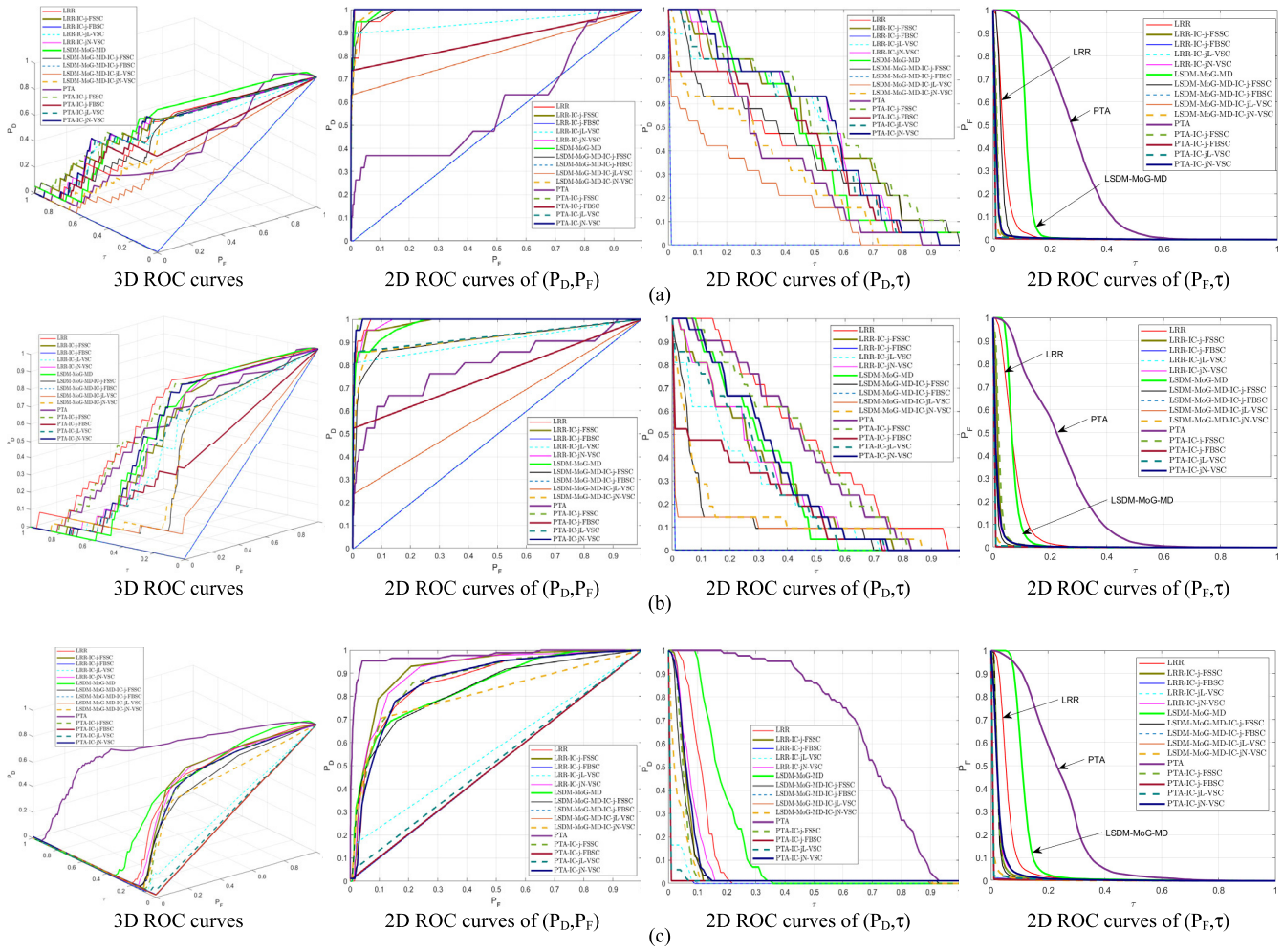



Fig. 17. 3-D ROC curves along with their corresponding 2-D ROC curves produced by LRR, LSDM-MoG, and PTA with/without SC criteria for the HYDICE panel, HYDICE urban, and San Diego datasets. (a) HYDICE panel data. (b) HYDICE urban data. (c) San Diego data.

evaluation, specifically the 2-D ROC curve of $(P_F,\tau)$ to be used to evaluate the degree and level of BKG suppressibility that an anomaly detector can achieve. To see this, Fig. 17 plots the 3-D ROC curves along their corresponding 2-D ROC curves produced by Figs. 14–16 where PTA produced far worse BS highlighted by purple than any other anomaly detector, and LSDM-MoG-MD and LRR came as the second and third worst AD with BS highlighted by green and red, respectively, pointed out by arrows. These three plots clearly stood out from all other BS curves in Fig. 17 as the three most worst BS. However, according to [25], [27], [31], LRR, LSDM-MoG-MD, and PTA performed very well if we only looked into the results of 2-D ROC curves of $(P_D,P_F)$ as shown in Fig. 17. These experiments further demonstrated that these three most recently developed anomaly detectors, LRR, LSDM-MoG-MD, and PTA, did very poorly in BS even though they did well in TD. In other words, the commonly used traditional 2-D ROC curves of $(P_D,P_F)$ to evaluate detection performance are misleading and not reliable.

The truth of matter is that how much BS can be accomplished is really determined by BKG and noises. Apparently, LRR, LSDM-MoG-MD, and PTA could not address these two issues effectively.

1) PTA decomposed the data space into a BKG tensor and an anomaly tensor for hyperspectral AD so that the anomalies can be detected in the anomaly tensor in terms of TD. Unfortunately, PTA could not generate an individual noise tensor to be separated from BKG and anomaly tensors since noise is random and cannot be characterized by sparse signals. This results in very poor performance of PTA in BS.
2) On the other hand, there are three issues associated with LSDM-MoG-MD in [25] which modeled BKG by a mixture of Gaussians. One is how many Gaussians are sufficient to describe BKG. This is because a linear mixture of Gaussians is still Gaussian. A second issue is not all the BKG signal sources can be characterized by Gaussians. A third issue is how to deal with non-Gaussian noises which are high-order statistics and cannot be modeled by Gaussians.
3) The LRR-based AD approaches also suffer from similar issues as well where the BKG dictionary cannot fully characterize and represent BKG, while sparse representation by regularization can only constrain the spectral components not the entire 3-D data cube. The EAS proposed in Section IV can resolve this issue using

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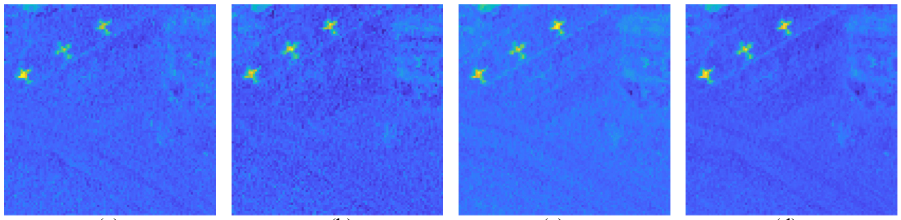



Fig. 18. Detection maps of CEM using (a) $\mathbf{d}_1$, (b) $\mathbf{d}_2$, (c) $\mathbf{d}_3$, and (d) $\bar{\mathbf{d}}$ as desired target signatures.

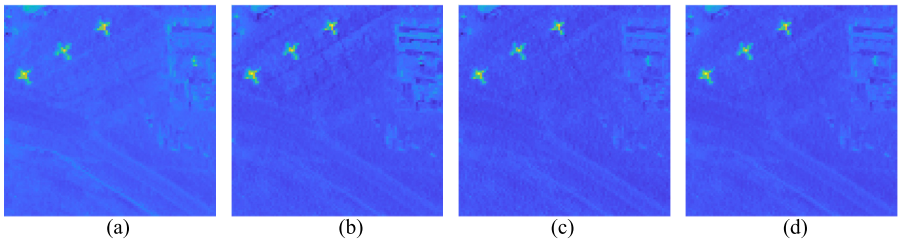



Fig. 19. Detection maps of KCEM using (a) $\mathbf{d}_1$, (b) $\mathbf{d}_2$, (c) $\mathbf{d}_3$, and (d) $\bar{\mathbf{d}}$ as desired target signatures.

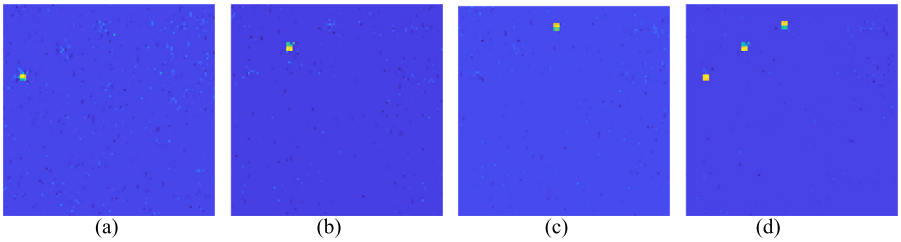



Fig. 20. Detection maps of hCEM using (a) $\mathbf{d}_1$, (b) $\mathbf{d}_2$, (c) $\mathbf{d}_3$, and (d) $\bar{\mathbf{d}}$ as desired target signatures.

$k$-VSC with $k = jL$ or $jN$ which takes care of a 3D data cube. Interestingly, the idea of VSC presented in SectIV may provide a new idea to extend LRaSR-model based methods by constraining a sparse representation in 3-D data cube instead of spectral vectors as LRR in [27].

A final remark is worthwhile. Despite that BKG suppressibility of LRR, LSDM-MoG-MD, and PTA can be improved by EAS, the experiments in Section VI did show that their performance using EAS still could not compete against IC using $k$-VSC and CDA using $k$-VSC with $k = jL$ or $jN$.

## VIII. Issue of Using San Diego Airport Data Scene for Anomaly Detection

As noted in the experiments of Sections VI-B.3 and VII-C, there is an issue of using the San Diego airport data scene for AD because many currently developed and existing anomaly detectors produced very high $\text{AUC}_{(\text{D,F})}$ values for this scene and then claimed to perform better than other anomaly detectors. Unfortunately, as shown in the discussions for the experiments of this particular scene in Sections VI-B.3 and VII-C, the conclusions solely based on TD without taking BS into account were misleading. As a matter of fact, according to the eight 3-D ROC curve-derived detection measures in Table XI, these methods indeed performed poorly BS by producing very high $\text{AUC}_{(\text{F},\tau)}$ values. This section conducts an interesting study on the issue of using this scene for AD.

According to the ground truth provided by Fig. 7(c), the anomalies are three airplanes which are very large and clearly visible by inspection. Intuitively speaking, a target is called an anomaly because the target cannot be known by visual inspection or prior knowledge as described in the introduction. With this interpretation, these airplanes should not be considered as anomalies because they can be identified clearly by visual inspection. As shown in Fig. 7(b), this scene should have many anomalies present at the right upper corner. Without ground truth, they cannot be identified. This is exactly what targets we are interested as anomalies detected in Fig. 16 where the anomaly detectors should be evaluated based on these detected anomalies, not the three airplanes.

In the following experiments, we will demonstrate that by visual inspection of the three airplanes in this scene, a very few data samples from the airplanes can be collected as training samples to be used as target knowledge. So, three pixels, one for each of the centers of the three airplanes, were selected as targets of interest. Then, a $3 \times 3$ window centered at each selected pixel was used to calculate the desired target signature to represent each of the airplanes, $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{d}_3$. In addition, the average of $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{d}_3$ was also calculated by $\bar{\mathbf{d}} = (1/3)(\mathbf{d}_1 + \mathbf{d}_2 + \mathbf{d}_3)$ to represent the desired target signature for these three airplanes. Then we implemented four CEM-based detectors, CEM [8], [63]–[65], kernel CEM

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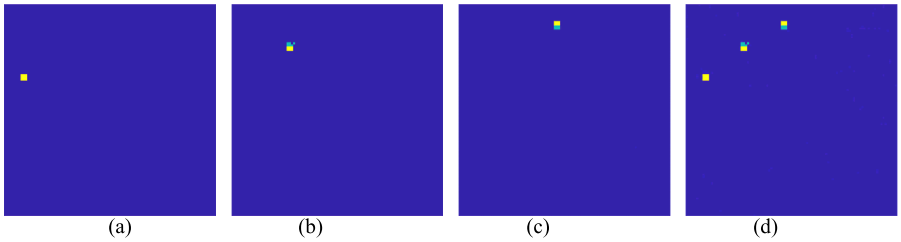



Fig. 21. Detection maps of ECEM using (a) $\mathbf{d}_1$, (b) $\mathbf{d}_2$, (c) $\mathbf{d}_3$, and (d) $\bar{\mathbf{d}}$ as desired target signatures.

(KCEM) [66], hierarchical CEM (hCEM) [67], and ensemble cascaded CEM (ECEM) [68], for target detection using $\mathbf{d}_1$, $\mathbf{d}_2$, $\mathbf{d}_3$, and $\bar{\mathbf{d}}$ as the desired target signatures to detect the three airplanes. Figs. 18–21 show their detection maps, respectively, where the brightest pixels in all the figures in Figs. 18–21 are those pixels selected as training samples to calculate the desired target signatures. Interestingly, CEM in Fig. 18 and KCEM in Fig. 19 detected all the three airplanes regardless of which signatures, $\mathbf{d}_1$, $\mathbf{d}_2$, $\mathbf{d}_3$, and $\bar{\mathbf{d}}$, were being used for the target signature while KCEM achieving much better BS in Fig. 19. This indicated that all the three airplanes are indeed considered as the same type of targets. In contrast, hCEM and ECEM in Figs. 20 and 21 show otherwise where $\mathbf{d}_1$, $\mathbf{d}_2$, $\mathbf{d}_3$, and $\bar{\mathbf{d}}$ could actually differentiate the three airplanes very well. Unfortunately, since hCEM and ECEM were so sensitive to the used target signatures, only those pixels selected to calculate the target signatures could be detected, while all the other pixels in the three airplanes could not. If we further compare the results in Figs. 18–21 with the results in Fig. 16, specifically those produced by PTA, the target detection by CEM and KCEM certainly performed significantly better than AD by RX-AD, CEM-AD, and CDA-AD which used the models of LRR, LSDM-MoG, and PTA. This is mainly due to the fact that anomaly detectors could have used the very little partial prior knowledge of the airplanes obtained by visual inspection but did not take advantage of it. These experiments provided hard evidence that constructing EAS is crucial.

The experiments presented in this section clearly showed that using the San Diego airport scene for AD is very likely to result in misleading conclusions. This is because this scene generally provides very good TD with very large target airplanes as anomalies but very poor BS in AD as demonstrated in [9], [32], [33], [47], [48], and [69]. The same conclusion is also applied to other similar scenes such as airport-urban-beach (ABU) data scene available on http://xudongkang.weebly.com.

As a concluding remark, in order for the San Diego airport data scene to be considered for AD, we should include the entire scene in Fig. 7(a) where the airplanes will look more like anomalies not visible large targets in Fig. 7(b). However, this whole scene is too large to be used for data processing. This may explain the reason why the cropped scene in Fig. 7(b) has been widely used for AD instead. Unfortunately, this turns out to be not a good idea as shown by the above experiments.

## IX. Conclusion

The effectiveness of AD is generally determined by two key factors, BKG suppressibility and anomaly detectability. The fact of matter is that anomalies are generally embedded in BKG and also contaminated by noise. Over the past years, AD has been mainly focused on how to cope with the BKG issue but not on how to enhance anomaly detectability. This has been witnessed by many low-rank and sparse space/representation-based models recently developed in the literature, all of which attempted to decompose or separate the data space into two spaces, one representing a low-rank space corresponding to BKG and the other representing a sparse space corresponding to AS. Unfortunately, once the sparse space is separated, very little has been done with it. This article defines a new concept of EAS and further develops methods of how to construct EAS. It first makes use of DS to removing 2OS that are generally used to characterize BKG and then uses FastICA to generate ICs that can be used to form an AS for AD. However, there are three issues that arise in such AS. One is the use of random initial conditions by FastICA which results in nonrepeatable ICs. A second one is its inability in separating HOS non-Gaussian noises/interferers from AS. This is a critical issue because hyperspectral image noise is generally non-Gaussian. A third one is how to select appropriate $j$ ICs to form AS, $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$. The new defined EAS is a result of addressing these three issues. The construction of EAS first uses ATGP to not only specify initial condition for each FastICA-generated IC to resolve the IC nonrepeatability issue but also to prioritize its generated ICs. It then develops four versions of SC to be imposed on $\mathbf{X}^{\mathrm{AS}}_{\mathbf{IC}^j}$ to remove non-Gaussian noises/interferers. Furthermore, it takes advantage of VD and MXSVD to automatically determine the value of $j$. In other words, $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ can be constructed by combining ATGP-FastICA, SC, and VD/MXSVD. Once EAS is constructed, an anomaly detector can be implemented in EAS to improve AD performance. The novelty is that none of the above construction techniques of $\mathbf{X}^{\mathrm{EAS}}_{\mathbf{IC}^j}$ can be found in the ICA-based AD approaches. Most importantly, the four versions of SC derived from source coding in information theory have never been explored in any ICA-based approach and can be further investigated for future research in extending SC-constrained EAS to SC-constrained effective target space, SC-constrained effective endmember space, etc.

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## REFERENCES


[1] I. Racetin and A. Krtalić, "Systematic review of anomaly detection in hyperspectral remote sensing applications," *Appl. Sci.*, vol. 11, no. 11, p. 4878, May 2021, doi: 10.3390/app11114878.

[2] A. B. Nassif, M. A. Talib, Q. Nasir, and F. M. Dakalbab, "Machine learning for anomaly detection: A systematic review," *IEEE Access*, vol. 9, pp. 78658–78700, 2021, doi: 10.1109/ACCESS.2021.3083060.

[3] C.-I Chang, *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection*. New York, NY, USA: Springer, 2016.

[4] C.-I Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2022.

[5] C.-I Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[6] S. I. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 1, pp. 1760–1770, Oct. 1990.

[7] C.-I Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[8] C.-I Chang, *Hyperspectral Imaging: Techniques for Spectral detection and Classification*. Norwell, MA, USA: Kluwer, 2003.

[9] C.-I Chang, "Constrained energy minimization anomaly detection for hyperspectral imagery via dummy variable trick," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2022.

[10] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[11] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "A locally adaptive background density estimator: An evolution for RX-based anomaly detectors," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 323–327, Jan. 2014.

[12] A. Banerjee, P. Burlina, and C. Diehl, "A support vector method for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2282–2291, Aug. 2006.

[13] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RX-AD and linear filter-based RX-AD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[14] C.-I Chang, Y. Wang, and S.-Y. Chen, "Anomaly detection using causal sliding windows," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 7, pp. 3260–3270, Jul. 2015.

[15] H. Kwon, S. Z. Der, and N. M. Nasrabadi, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, Nov. 2003.

[16] L. Wei and D. Qian, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[17] W. Liu and C.-I Chang, "Multiple window anomaly detection for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 658–664, Apr. 2013.

[18] W. Xie, T. Jiang, Y. Li, X. Jia, and J. Lei, "Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4218–4230, Jul. 2019.

[19] J. Lei, W. Xie, J. Yang, Y. Li, and C.-I Chang, "Spectral–spatial feature extraction for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8131–8143, Oct. 2019.

[20] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[21] Y. Wang *et al.*, "A posteriori hyperspectral anomaly detection for unlabeled classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3091–3106, Jun. 2018.

[22] S.-Y. Chen, Y. Wang, C.-C. Wu, C. Liu, and C.-I Chang, "Real-time causal processing of anomaly detection for hyperspectral imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 2, pp. 1511–1534, Apr. 2014.

[23] S. Chang, B. Du, and L. Zhang, "BASO: A background-anomaly component projection and separation optimized filter for anomaly detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 3747–3761, Jul. 2018.

[24] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[25] L. Li, W. Li, Q. Du, and R. Tao, "Low-rank and sparse decomposition with mixture of Gaussian for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 9, pp. 4363–4372, Sep. 2021.

[26] D. Chong, B. Hu, X. Gao, H. Gao, P. Xia, and Y. Wu, "Hyperspectral deep convolution anomaly detection based on weight adjustment strategy," *Appl. Opt.*, vol. 59, no. 31, pp. 9633–9642, 2020.

[27] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[28] E. J. Candes, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *J. ACM*, vol. 58, no. 3, pp. 1027–1063, 2009.

[29] T. Zhou and D. Tao, "GoDec: Randomized low-rank and sparsity matrix decomposition in noisy case," *Proc. 28th Int. Conf. Mach. Learn.*, Bellevue, WA, USA, 2011, pp. 1–16.

[30] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1389–11376, Mar. 2016.

[31] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[32] C.-I Chang, H. Cao, S. Chen, X. Shang, M. Song, and C. Yu, "Orthogonal subspace projection-based GoDec for low rank and sparsity matrix decomposition for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2403–2429, Mar. 2021.

[33] S. Chen, C.-I Chang, and X. Li, "Component decomposition analysis for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–22, 2022.

[34] A. Hyvarinen, J. Karhunen, and E. Oja, *Independent Component Analysis*. Hoboken, NJ, USA: Wiley, 2001.

[35] A. Hyvärinen and E. Oja, "A fast fixed-point for independent component analysis," *Neural Comput.*, vol. 9, no. 7, pp. 1483–1492, 1997.

[36] A. Hyvarinen, "Fast and robust fixed-point algorithms for independent component analysis," *IEEE Trans. Neural Netw.*, vol. 10, no. 3, pp. 626–634, May 1999.

[37] A. Hyvärinen and E. Oja, "Independent component analysis: Algorithms and applications," *Neural Netw.*, vol. 13, nos. 4–5, pp. 411–430, Jun. 2000.

[38] H. Ren and C.-I Chang, "Automatic spectral target recognition in hyperspectral imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 4, pp. 1232–1249, Oct. 2003.

[39] J. Wang and C.-I Chang, "Independent component analysis-based dimensionality reduction with applications in hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1586–1600, Jun. 2006.

[40] C.-I Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[41] C.-I Chang, *Virtual Dimensionality for Hyperspectral Imagery*. Bellingham, WA, USA: SPIE, Sep. 2009, doi: 10.1117/2.1200909.1749.

[42] C.-I Chang, W. Xiong, H.-M. Chen, and J.-W. Chai, "Maximum orthogonal subspace projection approach to estimating the number of spectral signal sources in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 504–520, Jun. 2011.

[43] C.-I Chang, W. Xiong, and C.-H. Wen, "A theory of high-order statistics-based virtual dimensionality for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 188–208, Jan. 2014.

[44] C.-I Chang, "A review of virtual dimensionality for hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1285–1305, Apr. 2018.

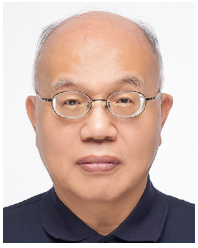
[45] O. Kuybeda, D. Malah, and M. Barzohar, "Rank estimation and redundancy reduction of high-dimensional noisy signals with preservation of rare vectors," *IEEE Trans. Signal Process.*, vol. 55, no. 12, pp. 5579–5592, Dec. 2007.

[46] R. J. McEliece, *Theory of Information and Coding*. New York, NY, USA: Cambridge Univ. Press, 2004.

[47] C.-I Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.


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[48] C.-I Chang, H. Cao, and M. Song, "Orthogonal subspace projection target detector for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4915–4932, 2021.
[49] N. Vaswani, T. Bouwmans, S. Javed, and P. Narayanamurthy, "Robust subspace learning: Robust PCA, robust subspace tracking, and robust subspace recovery," *IEEE Signal Process. Mag.*, vol. 35, no. 4, pp. 32–55, Jul. 2018.
[50] K. C. Tiwari, M. K. Arora, D. Singh, and D. Yadav, "Military target detection using spectrally modeled algorithms and independent component analysis," *Opt. Eng.*, vol. 52, no. 2, Feb. 2013, Art. no. 026402.
[51] J. A. Jablonski, T. J. Bihl, and K. W. Bauer, "Principal component reconstruction error for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1725–1729, Aug. 2015.
[52] R. J. Johnson, J. P. Williams, and K. W. Bauer, "AutoGAD: An improved ICA-based hyperspectral anomaly detection algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 6, pp. 3492–3503, Jun. 2013.
[53] S. Song, H. Zhou, H. Qin, K. Qian, K. Cheng, and J. Qian, "Hyperspectral image anomaly detecting based on kernel independent component analysis," *Proc. SPIE*, vol. 10697, Feb. 2018, Art. no. 1069710, doi: 10.1117/12.2309936.
[54] S. Song, H. Zhou, J. Zhou, K. Qian, K. Cheng, and Z. Zhang, "Hyperspectral anomaly detection based on anomalous component extraction framework," *Infr. Phys. Technol.*, vol. 96, pp. 340–350, Jan. 2019.
[55] D.-M. Tsai and S.-C. Lai, "Independent component analysis-based background subtraction for indoor surveillance," *IEEE Trans. Image Process.*, vol. 18, no. 1, pp. 158–167, Jan. 2009.
[56] C. Guo and M. Zhu, "Indoor video moving objects detection based on the independent component analysis with reference," in *Proc. Int. Congr. Signal Process. (CISP)*, Nov. 2010, pp. 322–326.
[57] P. K. Varshney and M. K. Arora, *Advanced Image Processing Techniques for Remotely Sensed Hyperspectral Data*. New York, NY, USA: Springer-Verlag, 2004.
[58] C.-I Chang, *Hyperspectral Data Processing: Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.
[59] J. Wang and C.-I Chang, "Applications of independent component analysis in endmember extraction and abundance quantification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 9, pp. 2601–2616, Sep. 2006.
[60] C.-I Chang, S.-S. Chiang, J. A. Smith, and I. W. Ginsberg, "Linear spectral random mixture analysis for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 2, pp. 375–392, Feb. 2002.
[61] H. V. Poor, *An Introduction to Signal Detection and Estimation*, 2nd ed. New York, NY, USA: Springer-Verlag, 1991.
[62] C.-I Chang, *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection*. New York, NY, USA: Springer, 2016.
[63] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Elect. Eng., Univ. Maryland, Baltimore County, MD, USA, Aug. 1993.
[64] W. Farrand, "Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.
[65] C.-I Chang, "Target signature-constrained mixed pixel classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 5, pp. 1065–1081, May 2002.
[66] K. Y. Ma and C.-I Chang, "Kernel-based constrained energy minimization for hyperspectral mixed pixel classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2022.
[67] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.
[68] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, p. 1310, Jan. 2019, doi: 10.3390/rs1111131.
[69] C.-I Chang and J. Chen, "Hyperspectral anomaly detection by data sphering and sparsity density peaks," *IEEE Trans. Geosci. Remote Sens.*, early access, Mar. 22, 2022, doi: 10.1109/TGRS.2022.3161462.



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